

AKADEMIKA

DCL-03
PULSE CODE MODULATION
TRANSMITTER KIT

DCL-04
PULSE CODE DEMODULATION
RECEIVER KIT

EXPERIMENTAL MANUAL

....A MESSAGE FROM

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It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

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As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that its integrity is checked with respect to its checklist. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In DCL-03 and DCL-04 kits, signals are to be monitored with an oscilloscope, or with the help of indicators as explained in the manual. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

These kits must be employed only for the use for which it has been conceived, i.e. as educational kits, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and upgradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

Input Channels	2 numbers, which are time division multiplexed and Pulse Code modulated.										
Onboard Signals	Two variable amplitude synchronized sine waves of Frequency 1 KHz and 500 Hz Two variable amplitude DC levels										
Frame Synchronization	By Pseudo Random Sequence generator/detector circuits.										
Bit Synchronization	By Phase Locked Loop techniques.										
Mode Of Operation	Fast / Slow modes.										
Serial Data Rate	Fast Mode: 240 bits per second. Slow Mode: 1 bit per 813 milliseconds and 1 bit per 1.6 second.										
Sampling Frequency	Fast Mode: 16 KHz Slow Mode: 1 sample per 11 seconds and 1 sample per 22 seconds										
Error Control Code Facility	Even, Odd, Hamming, None.										
Switch Faults	8 Switch Faults on DCL-03 plus 4 Switch Faults on DCL-04										
Channel Data Patterns	<table> <tr> <th><u>Pattern</u></th><th><u>Error control option</u></th></tr> <tr> <td>7 bit data</td><td>NONE</td></tr> <tr> <td>6 bit data + 1 bit odd parity</td><td>ODD</td></tr> <tr> <td>6 bit data + 1 bit even parity</td><td>EVEN</td></tr> <tr> <td>4 bit data + 3 bit error check code</td><td>Hamming</td></tr> </table>	<u>Pattern</u>	<u>Error control option</u>	7 bit data	NONE	6 bit data + 1 bit odd parity	ODD	6 bit data + 1 bit even parity	EVEN	4 bit data + 3 bit error check code	Hamming
<u>Pattern</u>	<u>Error control option</u>										
7 bit data	NONE										
6 bit data + 1 bit odd parity	ODD										
6 bit data + 1 bit even parity	EVEN										
4 bit data + 3 bit error check code	Hamming										
Serial Data Patterns	14-bit pseudo random sequence plus 14 bits of data from two channels										
Connection Details	Sockets and connecting chords are provided for connections on board and for connection from one board to another.										
Test Points	All relevant test points are brought out for observation.										
Visual Indications	LED's are provided onboard, for observation of data and control flows.										
Power Supply	-12V, +12V, +5V, GND										
Accessories											

DCL -03

Quantity

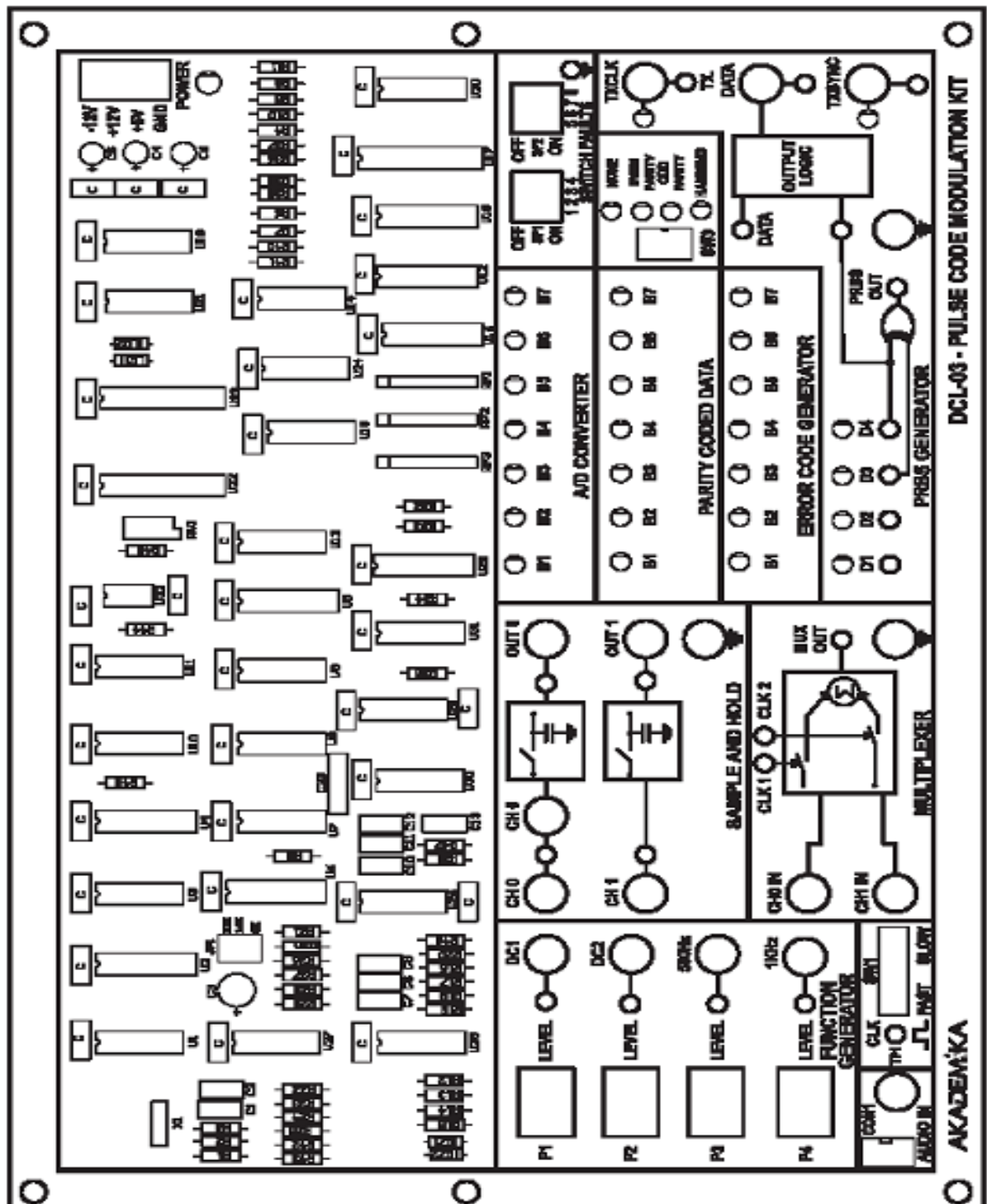
1	Mini Links PCS2 Patch cord	Male To Male	6
2	DCL-03 & DCL-04 EXPT Manual		1
3	Power Supply		1
4	Power cord		1

DCL -04

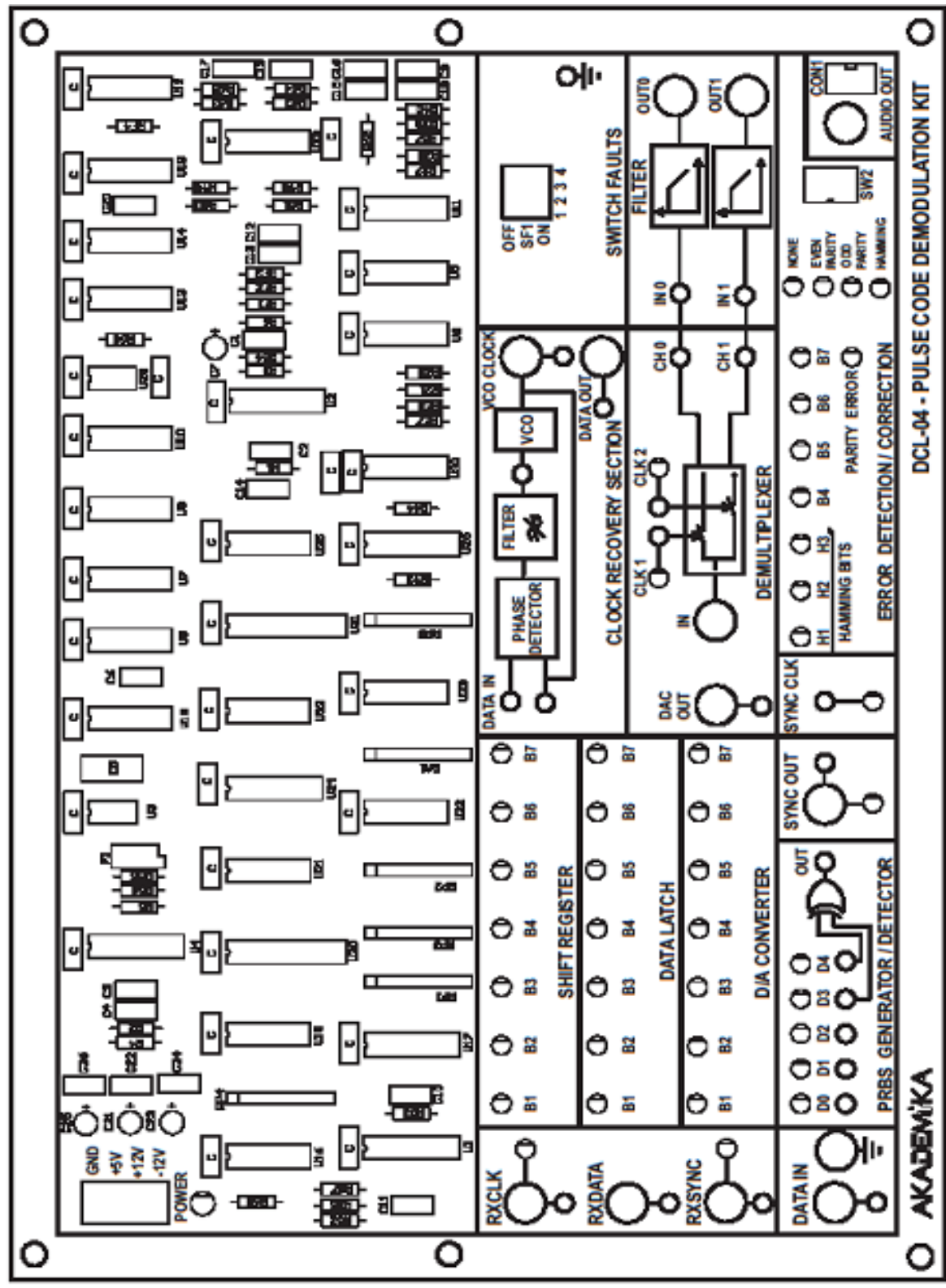
1	Mini Links PCS2 Patch cord	Male To Male	2
2	Short Links PCS2 Patch cord	Male To Male	5

FUNCTIONAL BLOCK

DCL-03



DCL-04



INTRODUCTION

Akademika with the set of two trainers, namely, “DCL-03” and “DCL-04” initiates the user to understand the fundamentals of Pulse Code Modulation Techniques.

The various principles of Pulse Code Modulation, such as Sampling, Multiplexing, Quantizing and Encoding, can be studied with these trainer kits. The trainer also illustrates the use of parity bit for single bit error detection and Hamming Code for single bit error detection and correction. Frame Synchronization using a pseudo random bit sequence (PRBS) and bit Synchronization using phase locked loops are also illustrated in these kits.

System Features

- 2-Channel Time Division Multiplexed PCM Transmission / Reception. Each channel data is coded to a 7-bit data. Multiplexed with this 14-bit data from the two channels is a 14-bit pseudo random sequence for Synchronization purpose.
- The system offers two operating modes:
 - i] **Fast Mode:** The Fast Mode is for real time operation, where various signals can be observed on an oscilloscope. In this mode, the serial data transmission frequency is 240 KHz. The sampling frequency per channel is 16 KHz.
 - ii] **Slow Mode:** The slow Mode is for examination of control and data flow on LED's. In this, sampling rate per channel is 11 seconds. Serial data transmission rate is 813 milliseconds. Also we can reduce sampling rate per channel to 22 seconds.
- Transmitter includes an onboard function generator to generate two synchronized, amplitude variable sine waves (1 KHz and 500 Hz), for use as analog inputs in Fast Mode and two amplitude variable DC levels for the use as analog inputs both in Fast Mode and Slow Mode.
- Operations with the following error check code options are available:
 - i] **None:** Here each sample is encoded as a 7-bit data word. In this mode no error detection or correction facilities are available. In this case any error goes unnoticed.
 - ii] **Even Parity:** In this mode of operation, each sample is encoded as a 6-bit data plus an even parity bit. A single bit error occurring in the transmission can be detected at the receiver.
 - iii] **Odd Parity:** In this mode of operation, each sample is encoded as a 6-bit data plus an odd parity bit. A single bit error occurring in the transmission can be detected at the receiver.
 - iv] **Hamming Code:** In this mode of operation, each sample is encoded as a 4-bit data plus 3-bit error check code. A single bit error occurring in the transmission can be detected and corrected at the receiver.

One of the four modes of operations discussed above can be selected using the DIP switch provided.
- Single bit error conditions are simulated with the help of the DIP switch provided. These allow us to forcibly introduce an error in any one of the 4 most significant bit positions of the channel data bits.
- 8 Switch Faults on DCL-03 and 4 Switch Faults on DCL-04 are provided onboard to study different effects of faults introduced in the circuit.
- Facilities are provided for studying the Synchronisation between Transmitter and Receiver in 3 stages:
 - i] Direct hard-wired SYNC, CLOCK and DATA connections between Transmitter and Receiver in both FAST and SLOW mode of operations.
 - ii] Direct hard-wired CLOCK and DATA connections in FAST and SLOW mode of operation. The pseudo random sequence (PRBS) is used for deriving the SYNC signal at the Receiver.

iii] Direct hard-wired DATA connection only. Here CLOCK is derived using the phased locked loop at the Receiver. SYNC signal is generated by the pseudo random sequence (PRBS) generator/detector at the Receiver. This feature is offered under FAST mode of operation only.

CIRCUIT DESCRIPTION

- **Function Generator**

Transmitter Timing Logic two different frequencies say 16 KHz & 8 KHz are taken. This signal is fed to serial to parallel shift register U25, U27 (IC 74HC164), which generates the sine wave, by serial shift operations. The serial to parallel shift register (IC 74HC164) has resistive ladder network at the output. For 16 shifts of the register, one stair case sine wave is produced. So if a 16 KHz clock is fed to the shift register, 1KHz sine wave is generated and similarly 500Hz sine wave from 8 KHz with amplitude variable from 0-5 Volts. Two amplitude variable DC levels are also generated which are used as analog information signals for both FAST and SLOW mode operations.

- **Clock Generator**

This block consists of Crystal Oscillator, which generates a 6.4 MHz clock, from IC 74LS04 and supplied to U2 (IC 74LS390) decade counter from which all the transmitter data and timing signals are derived.

- **Transmitter Timing Logic**

This Logic generates the various timing signals for the Transmitter. Two synchronized amplitude variable sine waves are generated which are used as analog information signals for fast mode operation. The 6.4 MHz crystal oscillators generate a 6.4 MHz clock. It is divided by 2-decade counters U2 (IC 74LS390) and ripple counter U3 (IC 74LS393) to get the 32 KHz, 16 KHz, 8 KHz and 4 KHz frequencies. This Logic also generates 240KHz Transmitter Clock with the help of two input mux U6 (IC 74LS157) and JK flip flop U11 (IC 74LS74) 16KHz Sampling Clock for FAST mode operations, and 1.23Hz & 0.6Hz Transmitter clock and 0.088Hz & 0.044Hz Sampling Clock for SLOW mode operations. A Frame Sync Signal is also generated by U3, U10 and U11 this logic for Synchronization purposes.

- **Sample And Hold Logic**

This Logic samples the two information signals from the two analog inputs, using (IC 4016) U30.

- **Multiplexer Logic**

Two signals available at CH0 IN & CH1 IN are multiplexed their samples by interleaving them properly in their assigned time slots controlled by the timing signals TM-1 and TM-2 generated from U9 (IC 74LS74). The capacitor is provided at the output of the third buffer to provide a sample and hold mode to U30, which is a sampling switch CD4016.

- **Analog To Digital Converter (ADC)**

The Analog to Digital Converter converts the analog samples to digital bits. This device performs both the quantising and encoding operations. The analog to digital converter IC, AD673 forms the heart of this logic. AD673 is used for the analog to digital conversion of the multiplexed data. The timing of the A/D conversion is controlled by the convert pulse, which is generated in the transmitter timing logics. As soon as the data is ready, the AD673 gives a data ready (DR Active low) output. The data latch is used for latching the valid outputs of the A/D converter. The timing and latching interval of the data latch is

controlled by the latch enable signal U22 (74HCT374). Each sample is coded to a 7 bit data by the AD673.

- **Quantizing And Encoding**

Refer to the circuit diagram as shown later in this manual sheet 3 of 5 of **DCL-03**. The signal level of the input signal assumed to vary from 0 volt to 4.96 volts. The entire level is dividing to 128 uniform steps. Each step corresponds to

4.96V

----- = 40mV

128

The quantising level is chosen to be midway of the steps. If the signal level falls below the quantising level, then the signal is rounded OFF to the lower level. If the signal level falls above the quantising level, and then the signal is rounded OFF to the upper level. This type of quantising is called "uniform quantising, where the step levels are uniform. Then corresponding to the level chosen, code words are assigned to the samples. The code words vary from 0000000 to 1111111 for the 0 to 4.96 V. Thus the Analog to Digital converter assigns the code words for all the samples. For further understanding of the Analog to Digital Conversion principles, refer the data sheet of AD673.

- **Error Code Generator**

Refer to the circuit diagram as shown later in this manual sheet 2 & 4 of 5 of **DCL-03**. This block generates the Error Codes for detection and correction of single bit errors that occur in the transmission of the coded sequence.

The various Error Check Code options available are:

- * ODD Parity,
- * EVEN Parity, and
- * Hamming Code.

ODD PARITY: ODD Parity Generator generates the coded sequence, which consists of 6 bits of data plus one ODD parity Bit for transmission. The least significant bit of the data word from the ADC is replaced with the parity bit.

EVEN PARITY: EVEN Parity Generator generates the coded sequence, which consists of 6 bits of data pulse one EVEN Parity Bit for transmission. The least significant bit of the data word from the ADC is replaced with the parity bit.

HAMMING CODE: Hamming Code Generator generates the coded sequence, which consists of 4 bits of data plus 3 bits of error Check Bits for the transmission. The least significant 3 bits of the data word from the ADC are neglected.

NONE: Operation without any error check is also available, where the transmitted data consists of 7 bits of data only.

Anyone of the above four modes of operations can be selected, by using the switches provided onboard.

The ODD/EVEN parity generator U15 (IC 74LS280) generates odd parity and even parity bits for the data bits B2 through B7. But the parity bits are switched to output based on the control signals ODD parity control and EVEN parity control. When odd parity control goes high, odd parity bit is switched to the output and when even parity control goes high, even parity bit is switched to the output. Also note that hamming codes are generated only for the data bits B7 to B4.

- **Fault Generator**

This logic introduces single bit errors in any one of the bit position B4 to B7. When the switch SF2 is OFF, the data bit passes without inversion and when the switch is ON, the bit suffers an inversion, thus imposing a single bit error in the data stream. We can

observe that the 4-way DIP switch SF2 can force the single bit error in any of the four bit positions B4 to B7. Care should be taken not to switch more than one error in the data stream.

- **Error Check Code Introduction Logic**

In this logic, the error check codes are introduced at the least significant 3 bit positions. The control signals for this logic are generated by the use of a two-way DIP switch SF2. So when the switches are configured for NONE mode, the least significant 3 bits pass to the output without any alteration, when the switches are configured for ODD/EVEN parity mode the least significant bit is not transmitted and in that place, the ODD/EVEN parity bit is transmitted (based on the selection) and when the switches are configured for hamming mode, the least significant 3 bits are truncated and in their places the generated hamming codes are transmitted.

- **PRBS Generator**

For ensuring proper Frame Synchronization between the transmitter and the receiver, a 14-bit Pseudo Random sequence is transmitted along with the data. A 4-bit Shift Register U28 (IC74LS195) generates the Pseudo Random sequence, which is a unique pattern that has minimum correlation with the data. The clock frequency for the Pseudo Random Sequence Generator is the same as the frequency of the transmitted serial data. The PRBS generator is a 4-bit shift register with an EX-OR gate at the feedback. (The operation of PRBS generator is clearly illustrated in the experimental manual). Note that PRBS is generated and transmitted only when the sync remains high. When the sync is low, no PRBS is generated and at that time, the serial data is transmitted over the line.

- **Parallel To Serial Shift Register**

For shift register U12 (74LS165) converts the 7 bit per channel coded sequence at the output of the error control coding logic to a 7 bit serial data. From the two channels, we get 14 bit serial data per frame.

- **Transmitter Output Logic**

Refer to the circuit diagram as shown later in this manual sheet 2 of 5 of **DCL-03**. This logic transmits a serial data stream, which consists of 14 bit PRBS plus 14 bit data. The PRBS forms the header to the data. This output logic also ensures proper interleaving of data and PRBS in their respective time slots. Thus the Transmitter works and generates the serial bit stream.

Principles of Pulse Code Modulation Techniques

Pulse Code Modulation is a technique, where the samples are transmitted as coded words of finite bit (binary digit) length. In Pulse Code Modulation, first the samples are quantized, and then encoded before transmitting as a serial bit stream.

“Quantizing” is the process where the samples are made to assume one of the finite sets of discrete levels. Here, first the whole signal level is divided into a fixed number of discrete levels. The samples are rounded-off to the nearest discrete level. Then corresponding to the level chosen, a code word is assigned to the sample. This process is called “Encoding”. The process of quantizing and encoding together is called “Analog to Digital conversion”.

The parallel data word available after the analog to digital conversion is converted to a serial data stream after coding and sent through the channel. This coded data stream is said to be PCM coded data and is transmitted by the serially by the PCM transmitter.

For recovering the data from the serial data stream, first the serial data is converted to a parallel finite bit length code word. Then the receiver identifies the code word and assigns

the signal levels for the received code word. The functional block that performs this task of accepting sequences of binary digits and generating appropriate sequences of levels is called “Digital to Analog converter”. The sequences of level that appear at the output of the D/A converter is filtered to recover back the base band signal.

Information from more than one source can be sent in a common channel, by properly interleaving the samples from the various information sources in time domain. This process is called “Time Division Multiplexing”. At the Receiver, after the D/A conversion is over, a demultiplexing logic assigns the samples to the corresponding reconstruction units to recover the information signals.

Synchronization Techniques

For the recovery of information from the transmitted data stream, the transmitter and receiver should be perfectly synchronized. Synchronization between transmitter and receiver is achieved in two stages:

- 01) Bit Synchronization; and
- 02) Frame Synchronization.

Bit Synchronization

To recover the information from the serial bit stream, the receiver has to know, if the received data bit at a given instant of time, is ONE or ZERO. For determining the data bit, the clocks of transmitter and receiver are to be perfectly synchronized. Once this Synchronization is achieved, the receiver recovers the data, with reference to the clock.

For achieving this Synchronization, normally known as bit Synchronization, the clock of the transmitter (TX clock) can be sent along with the data. This calls for an additional channel for clock transmission. So normally a phase locked loop (PLL) is used at the Receiver for recovering the clock. The phase locked loop is made to lock with the incoming data whose VCO reproduces the clock at the Receiver.

Frame Synchronization

To recover the information transmitted from the data stream, the data is first converted to parallel code words. After the code words are decoded using the D/A converter, demultiplexing logic assigns the samples to the corresponding channel reconstruction units. For ensuring proper recovery of code words, and proper assignment of channel information's of the demultiplexing unit, the receiver has to know where the data begins exactly. For this, at the start of every frame, a 14-bit pseudo random sequence (PRBS) is sent, which forms the header of the data pattern. Subsequently what follows is data corresponding to the number of channels multiplexed.

The frame sync pulse can be sent by the Transmitter along with the data and clock to achieve frame Synchronization. But normally it is not sent, since it calls for an additional channel.

To recover the frame sync at the receiver from the data stream, a sequence similar to the one transmitted, is generated at the Receiver. The PRBS detector compares the generated sequence with the incoming data and locks whenever the incoming data matches with the generated sequence. The PRBS sequence is chosen in such a way that it has minimum correlation with the data to avoid false locks of the detector with the data bits. Once the PRBS detector is locked, it generates a frame sync pulse, whose rising edge corresponds to the start of the frame and the time to the slot when the pseudo random sequence occurs in the data stream. This frame sync pulse is used to code word and for demultiplexing purposes.

Pseudo Random Bit Sequences (PRBS)

Pseudo Random Bit Sequences are widely used for Frame Synchronization purpose. They form the header of the transmitted sequence. At the Receiver, these sequences are detected by a detector logic, which recovers the Synchronization signal at the receiver.

Pseudo Random Bit sequence, popularly known as Pseudo Random Code Signals, are actually deterministic signals. But they are called Pseudo Random Signals, because they have the statistical properties of Random Signals.

Example of Random Signal: Sample signal of white noise.

The Pseudo Random Signal has got an advantage that it has minimum correlation with any data signal, so that the false locking of the detector with data is avoided.

For a signal to be Pseudo Random in nature, it should satisfy the following properties of Randomness:

01) Balance Properties:

The given signal is said to possess good balance property, if the number of binary Ones differ from the number of binary Zeroes, by almost one digit in one period.

02) Run Property:

The Run is defined as a sequence of single type of binary digits (either binary Ones or binary Zeroes). The given sequence is said to satisfy the RUN property, if it satisfies the below condition:

Among the Runs of Ones and Zeroes in each period, it is desirable that about one half the Runs of each type are of length 1, about one fourth are of length 2, and one eighth are of length 3, and so on.

03) Correlation property:

If period of the sequence is compared term by term with any cyclic shift of itself, it is best if the number of agreements differs from the number disagreements by not more than 1.

In the DCL-03, PRBS sequence is generated by using a 4-bit right shift register whose feedback is completed by the EX-OR gate.

Let Initially 0001 be the stored nibble on the 4 bit shift registers:

Clock States	D1	D2	D3	D4	
			A	B	C
0	0	0	0	1	1
1	1	0	0	0	0
2	0	1	0	0	0
3	0	0	1	0	1
4	1	0	0	1	1
5	1	1	0	0	0
6	0	1	1	0	1
7	1	0	1	1	0
8	0	1	0	1	1
9	1	0	1	0	1
10	1	1	0	1	1
11	1	1	1	0	1
12	1	1	1	1	0
13	0	1	1	1	0
14	0	0	1	1	0
15	0	0	0	1	1
16	1	0	0	0	0

Thus the sequence repeats constantly with a period corresponding to 15 clock states.

Length of sequence = $2^4 - 1 = 15$

Now the pattern is 100010011010111

Check for Balance Property:

Number of ones in the sequence = 8

Number of zeros in the sequence = 7

Difference = 1

Hence the sequence is verified to have good balance property.

Check for Run Property:

Number of zero runs = 4

Number of runs of length 1 = 2 (half of 4)

Number of runs of length 2 = 1 (one fourth of 4)

The sequence is too short to go for further verifications like one eighth.

Thus the sequence is verified for run property with all ZERO runs. Similar verification can be done for the ONE run.

Check for Correlation Property:

Pseudo Random Sequence = 1 0 0 0 1 0 0 1 1 0 1 0 1 1 1

RRC Shifted Sequence = 1 1 0 0 0 1 0 0 1 1 0 1 0 1 1

a d a a d d a d a d d d a a

d = represents disagreements.

a = represents agreements.

RRC = Rotate Right with Carry

We see that:

Number of agreements = 7

Number of disagreements = 8

Difference = 1

Therefore the sequence satisfies correlation property also.

Thus the Pseudo Random Properties are verified for the Pseudo Random Sequence generated in "DCL - 03".

Note: Since the transmitted data contains only 14 bits of data, the generated PRBS is also restricted to have a 14-bit length. The last bit is truncated.

Error Control Coding Techniques

When the data is transmitted in the channel, bit errors may be introduced by noise and other factors existing in the channel. Error control coding techniques are used for detecting and correcting the errors. They can be used for controlling single bit, two bit, and three bit errors. Even Parity Codes, Odd Parity Codes, Hamming Codes are the normal error control coding techniques, used for the detection and correction of all single bit errors that occur in the transmission of data.

a) EVEN PARITY: EVEN Parity Generator generates the coded sequence, which consists of data stream plus one even parity bit. This Parity bit is added to the data word, such that the number of ones, in the coded sequence is EVEN.

Example:

Data	Parity Bit
111000	1
111100	0

At the Receiver, EVEN Parity Checker checks the number of Ones in the received coded sequences. If the number of Ones is EVEN, it indicates that no bit error had occurred during transmission. If the number of Ones in the received data stream is ODD, it is an indication of the single bit error that had occurred in the transmission of data. This method is effective only for the detection of single bit error. But the bit cannot be located and corrected. Also all two-bit errors go unnoticed in this check mode.

b) ODD PARITY: ODD Parity Generator generates the coded sequence, which consists of data stream plus one odd parity bit. This Parity is added to the data word, such that the number of ones, in the coded sequence is ODD.

Example:

Data	Parity Bit
111000	0
111100	1

At the Receiver, ODD Parity Checker checks the number of Ones in the received coded sequences. If the number of Ones is ODD, it indicates that no bit error had occurred during transmission. If the number of Ones in the received data stream is EVEN, it is an indication of the single bit error that had occurred in the transmission of data. This method is effective only for the detection of single bit error. But the bit cannot be located and corrected. Also all two-bit errors go unnoticed in this check mode.

c) HAMMING CODES: Hamming codes are the most effective single error detecting and correcting codes, used in practice. The hamming distance for such a code is 3, ensuring an effective correction of all single bit errors occurring in transmission:

The general format for Hamming Codes is as follows:

If N is the length of the coded sequence and n is number of error check bits

Then, $N = 2^n - 1$ (Where $n = 0, 1, 2, 3$)

Bit position of the check bits = 2^n

The typical size of some of Hamming Code normally used in practice is given below:

Size of the coded sequence	Size of Data	Size of Error Check bits
7	4	3
15	11	4
31	26	5
63	57	6

Thus depending upon the size of the coded sequence, hamming parity bits are generated. At the receiver the parity check is done and error bit is located and is corrected by bit reversal. Thus all single bit errors occurring in data transmission can be detected and corrected by this error check option.

Let us discuss in detail the (7,3) Hamming code, which is illustrated in "DCL - 03" and "DCL - 04":

Bit length of the coded sequence = 7

Number of error check bits = 3

Number of data bit = 4

In the coded sequence:

Let K1, K2, K3 be the error check bits.

L4, L5, L6, L7 be the data bits.

Now K1, K2, K3 are chosen in such a manner that

K1 - Even parity for the data bits L4, L5, L7

K2 - Even parity for the data bits L4, L6, L7.

K3 - Even parity for the data bits L5, L6, L7.

The K1, K2, K3 hamming bits and L4, L5, L6, and L7 are the data bits where L7 is the most significant data bit. Thus the coded sequence will be transmitted following format:

K1, K2, K3, L4, L5, L6, L7

L7 is shifted out first and K1 is the last.

Truth table for (7,3) Hamming Code is given below:

Data	Bit position
(In decimal)	

	K1	K2	K3	L4	L5	L6	L7
0	0	0	0	0	0	0	0
1	1	1	0	1	0	0	0
2	1	0	1	0	1	0	0
3	0	1	1	1	1	0	0
4	0	1	1	0	0	1	0
5	1	0	1	1	0	1	0
6	1	1	0	0	1	1	0
7	0	0	0	1	1	1	0
8	1	1	1	0	0	0	1
9	0	0	1	1	0	0	1
10	0	1	0	0	1	0	1
11	1	0	0	1	1	0	1
12	1	0	0	0	0	1	1
13	0	1	0	1	0	1	1
14	0	0	1	0	1	1	1
15	1	1	1	1	1	1	1

At the Receiver to detect and correct the error occurred transmission, EVEN Parity Check is done at bit positions.

Let K1*, K2*, K3* be the even parity check for the following bit of the coded sequence:

K1*: Even parity check bit for K1, L4, L5, L7 of the coded sequence

K2*: Even parity check bit for K2, L4, L6, L7 of the coded sequence

K3*: Even parity check bit for K3, L5, L6, L7 of the coded sequence

From the recovered K1*, K2*, and K3* the error bits are located and corrected.

Truth Table for error bit location is given below:

	K3*	K2*	K1*
No error	0	0	0
Error in data bit L4	0	1	1
Error in data bit L5	1	0	1
Error in data bit L6	1	1	0
Error in data bit L7	1	1	1

Once the error bits are located the error is corrected by bit reversal.

• Serial To Parallel Shift Register Logic

This logic takes care of the data word recovery from the Serial data stream. In the serial data stream, that is transmitted, only the valid data (of the two information channel) bits get shifted to parallel data from U19 (IC 74HC164) and are latched with U20 (IC 74HCT374). The PRBS is rejected. This is achieved by the control signals. RST164*, LE374*, OE374* that are generated in the receiver timing logic.

• Error Detect And Correct Logic

This offers the following options for detection and correction of all single bit errors that occur during the transmission.

• **Odd Parity Error Detector** This Logic detects the single bit error occurring in the data by ODD Parity Check. The Logic basically checks the number of ONE's in the data and determines if a single bit error had occurred during transmission. The indication is

given by PARITY ERROR LED for which output is coming from U31 PAL16V8. This Logic can neither locate the bit error position nor correct it.

Even Parity Error Detector

This Logic detects all single bit errors occurring in the data transmission by EVEN Parity Checks. The Logic checks the number of ONE's in the data stream and determines if a single bit error had occurred during transmission. The indication is given by PARITY ERROR LED whose output is coming from U31 GAL16V8. This Logic can neither locate the bit error positions nor correct it.

Thus the bit errors at the output of above two Logics goes uncorrected.

Hamming Error Detect and Correct Logic

This logic computes K1, K2, K3, at the receiver and locates the error bit position. From the received data bits, the 3 even parity generator generates the 3 control signal of the hamming code decoder namely H1, H2 and H3 in the under mentioned manner:

H1: Even parity bit for the bit positions: B1, B4, B5, B7.

H2: Even parity bit for the bit positions: B2, B4, B6, B7.

H3: Even parity bit for the bit positions: B3, B5, B6, B7.

Now when there is no error:

H1, H2, H3	=	0	0	0
		H3	H2	H1
Error in bit B7	=	1	1	1
Error in bit B6	=	1	1	0
Error in bit B5	=	1	0	1
Error in bit B4	=	0	1	1

When there is no error, outputs Y7, Y6, Y5, and Y3 of the U24 (IC 74LS138) remain low. When a single bit error occurs in the transmitted data, the corresponding output goes high indicating an error in the outputs and reverses and corrects the corresponding bit.

Once the error bit position is located, the error is corrected by bit reversal. Thus all single bit errors occurring in data transmission can be detected and corrected by this error check option.

NONE

The data can also be recovered without any error detect and correct options.

Any one of the above four modes of operations can be selected using the DIP switches.

- **Digital To Analog Converter** THE Logic converts the 7 bit coded sequence obtained at the output of the detect/correct logic to analog samples. This logic is built around the Digital to Analog Converter U26 (IC,DAC0800). It reproduces the multiplexed sample levels at the receiver.

When both "DCL -03" and "DCL -04" are set to NONE mode, the least significant 8 bits that are transmitted in the data stream are data bits. So they are passed directly to the D/A converter input. When both are set to ODD/EVEN parity mode, in the least significant bit that is transmitted is the parity bit, hence the LSB is truncated and it is forced to 0 before reaching D/A input. Instead, when both are set to hamming mode, the least

significant 3 bits that are transmitted are hamming bits and hence they are truncated and forced to 0's before reaching D/A input.

- **Pseudo Random Sync Detector**

The pseudo random sync generator/detector U17 (IC74LS195) recovers the frame sync pulse from the transmitted data and the clock. The clock is fed to the shift register input until the shift register output and the PRBS occurring in the data are perfectly matched, the whole system is reset and the sync lock output will be pulsating. When the whole system is locked, the sync lock output goes low forever and the Q output of the flip-flop U12 (74HCT74) directly gives the signal.

- **Receiver Timing Logic**

The Receiver Timing Logic produces the necessary timing signals for ensuring proper serial to parallel data conversion by U19 (IC 74HCT164). This conversion is of only the data bits, and not the PRBS demultiplexing signals to the two channels are also generated by this Logic.

- **Demultiplexing Logic**

Refer to the circuit diagram as shown later in this manual sheet 4 of 4 of **DCL-04**. The demultiplexing of the multiplexed sample levels is done by U6 (IC 4066) using this Logic, which assigns the samples of the two information sources to their corresponding reconstruction units.

- **Data Reconstruction Units**

Refer to the circuit diagram as shown later in this manual sheet 4 of 4 of **DCL-04**. It consists of Low Pass Butterworth Filters, which filters the high frequency components with the help of U33 (LM324) to recover back the base-band signals.

- **Bit Synchronization Circuits: (Clock Recovery System)**

Whenever the clock is not transmitted along with the data, then the bit synchronization circuit recovers the clock from the transmitted data for the receiver operations. The heart of this circuit is the digital phase locked loop, CD4046. The input signal to the PLL is the received data. The PLL tracks the incoming signal and goes for a lock only when the incoming signal frequency matches with the frequency of the voltage controlled oscillator (VCO) output. Once the PLL is locked, the receiver clock is recovered from the VCO of the PLL. Thus the bit synchronization logic recovers back the clock for the receiver operation.

Recovering information from serial data in PCM system needs the recovery of clock. Transmitting clock along with the data calls for an additional channel. So a system is build around phase locked loops to recover the clock, which is embedded in the data itself. Normal phase locked loop techniques fail for such an application, because the actual clock may appear for only short periods of time in the data for common encoding formats like NRZ. So a special logic is built to derive the clock from the data on such cases.

Principle of operation shows the data stream clock using a typical NRZ protocol. In this format, the ones are represented by highs and zeros are represented by the lows. In the data stream, it should be guaranteed those continuously more than 5 zeroes or 5 ones should not occur in succession while transmitting data. This ensures that the slowest rate of change will then be a frequency corresponding to one-twelfth clock. The fastest rate of change of frequency will correspond to one half of the clock frequency with alternate ones and zeroes occurring in the data stream. The block diagram of the clock recovery system, the 74LS123 (U1) and the OR gate 74LS32 (U11) form a bi-directional one shot that produces a positive output pulse for both rising and on its input. The period of this output pulses is set to one half, the total period of the clock, this is used to provide a frequency component in the data stream equal to clock even under worst case conditions of five

zeros and five ones, five zeroes etc. This can also be seen to double the frequency of the data stream, which is desirable as the PLL will now be able to lock to the original clock. The CD4046BE (U4) forms the PLL which when the actual clock appears in data equal to and produces a frequency at which its VCO output equal to and synchronized with the clock. The PLL's phase detector output is connected to the sample and hold amplifier input. It is also connected to the sample and hold output through a switch, this switch is held open and sample and hold amplifier is held in the sample mode as long as the clock appears in the data stream.

Whenever continuous ones or zeroes are present, the clock will not appear in the data stream and 74LS123 (U2) places the sample and hold amplifier in the hold mode and closes the switch. This holds the voltage of phase detector output and keeps proper driving voltage to the VCO, thus maintaining the frequency of the VCO output equals synchronized to the clock.

When the clock re-appears in the data stream, it drives the switch open and the amplifier to the sample mode, and the PLL once again track the clock in the data stream. The length of T1 is made equal to slightly less than the period of the clock, so that the S/H is always ready in the event of the clock it is not in the data stream and any sample to hold glitches will not be transmitted to the phase detectors output. The length of T2 is made slightly longer than the clock period, which will cause the switch to close immediately after one clock pulse is missed. These times T1 and T2 are set equal to $0.8T$ and $1.2T$ respectively.

- **Frame Synchronization**

Whenever the Frame sync Signal is not transmitted along with the data, the PRBS Generator/Detector circuit of the receiver regenerates the Frame Sync signal from the received clock and received data. At the receiver, a PRBS Generator U17 (74LS195) generates the 14-bit Pseudo Random Sequence, Which is similar to the PRBS that occur in the transmitted bit stream. The PRBS Detector compares the incoming bits with the generated Pseudo Random Sequence and locks when both match perfectly. The correlation between the Pseudo Random Sequences and the transmitted data is ensured to be minimum, so that the Frame Synchronization Circuit can reproduce the TX Sync Signal at the Receiver at the output of U12 (IC 74HCT74), which is an exact replace of the TX signal. Whenever there is a miss-match in the PRBS during a count, the system resets. The counter begins counting for the next status when the match recovers and hence maintains lock.

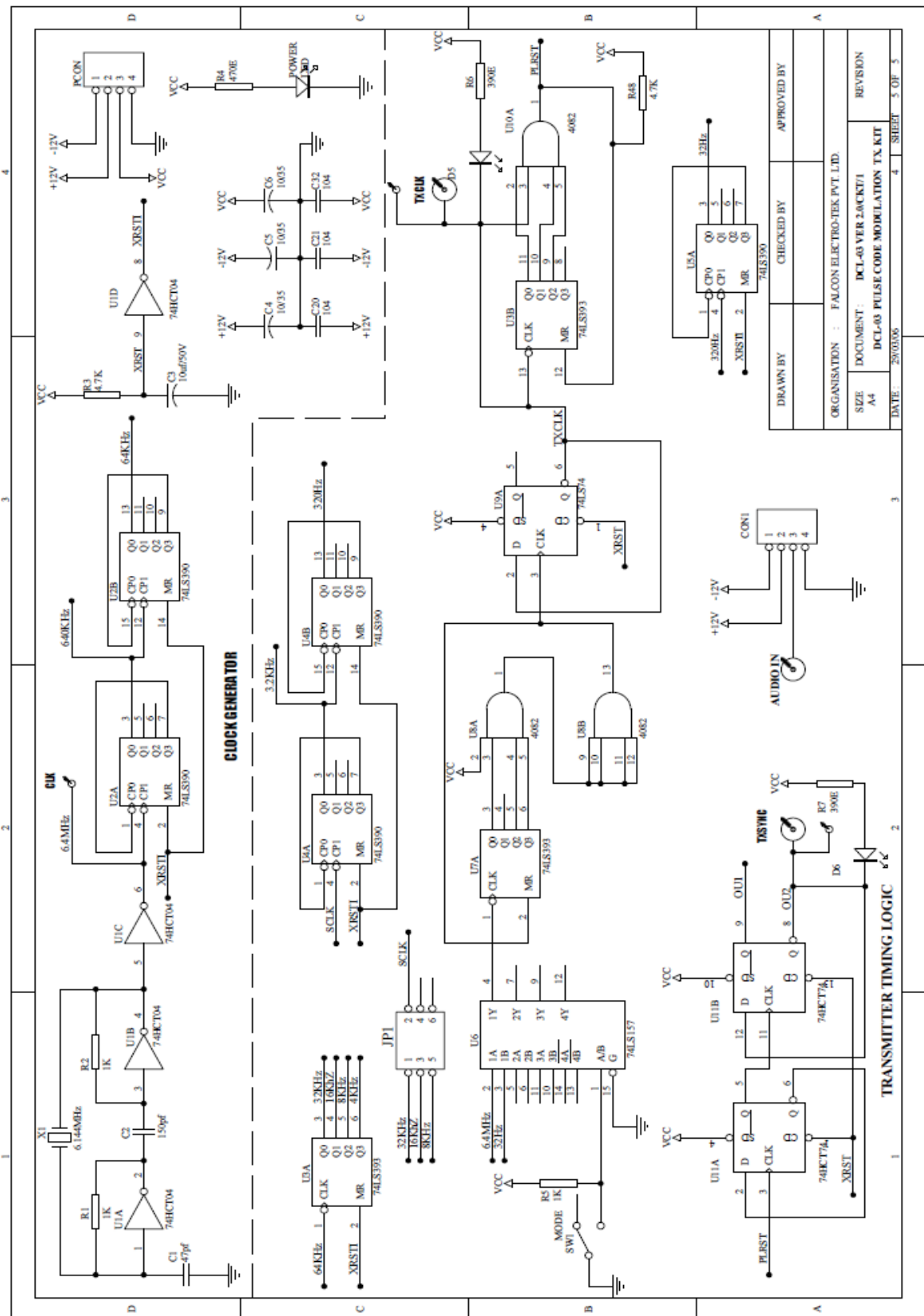
SWITCH FAULTS

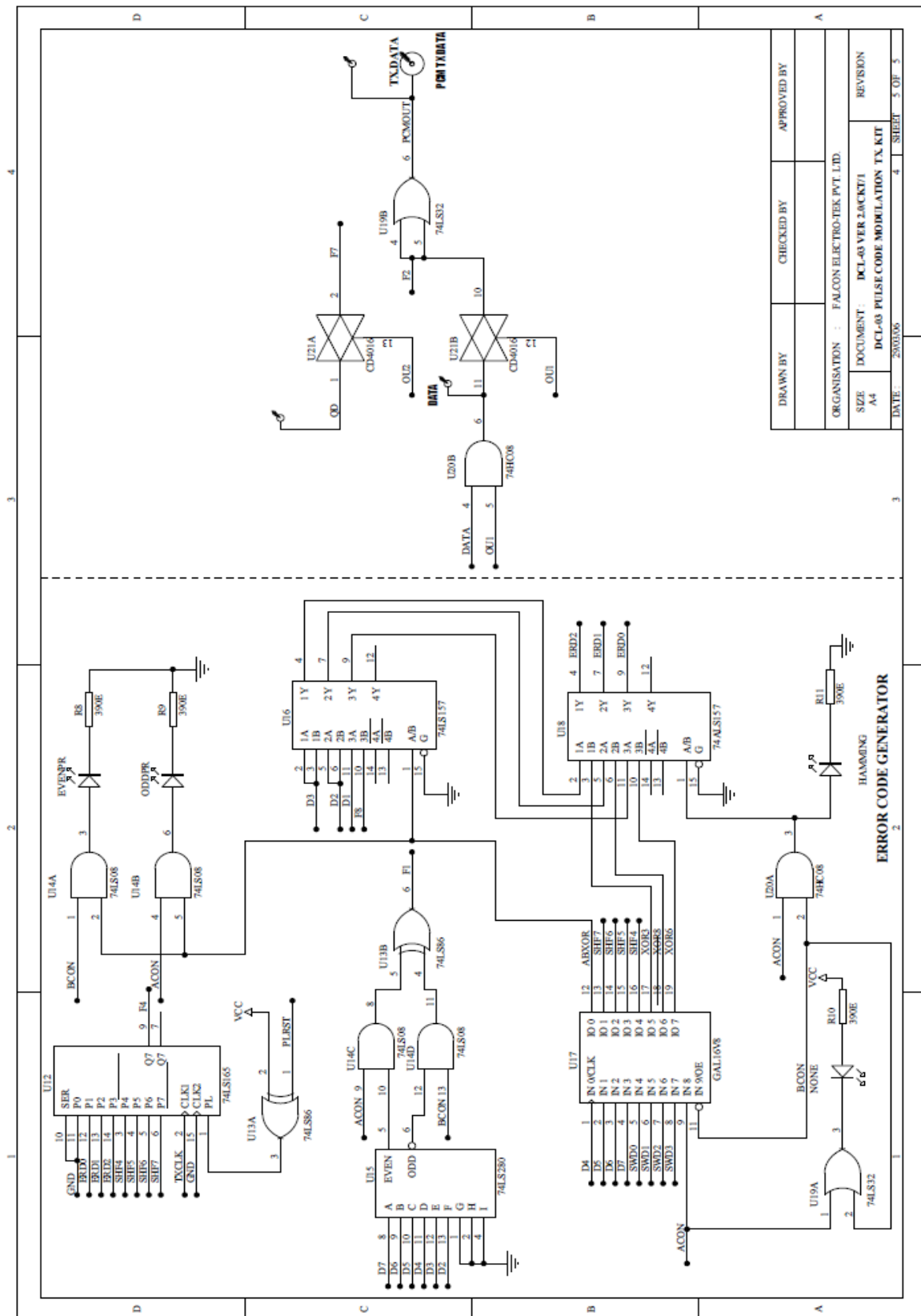
Switch fault section on DCL-03 is provided with 8 switches and switch fault section on DCL-04 is provided with 4 switches. These switches can be used to simulate fault conditions in various parts of the circuit. In DCL-03 first 4 faults are specially used only for introducing faults in data pattern (B4, B5, B6, B7) and remaining 4 faults are used to introduce faults in circuitry. The faults are normally used one at a time. To ensure that the DCL-03 kit and DCL-04 kit is fully operational, all switches should be set to OFF position before use.

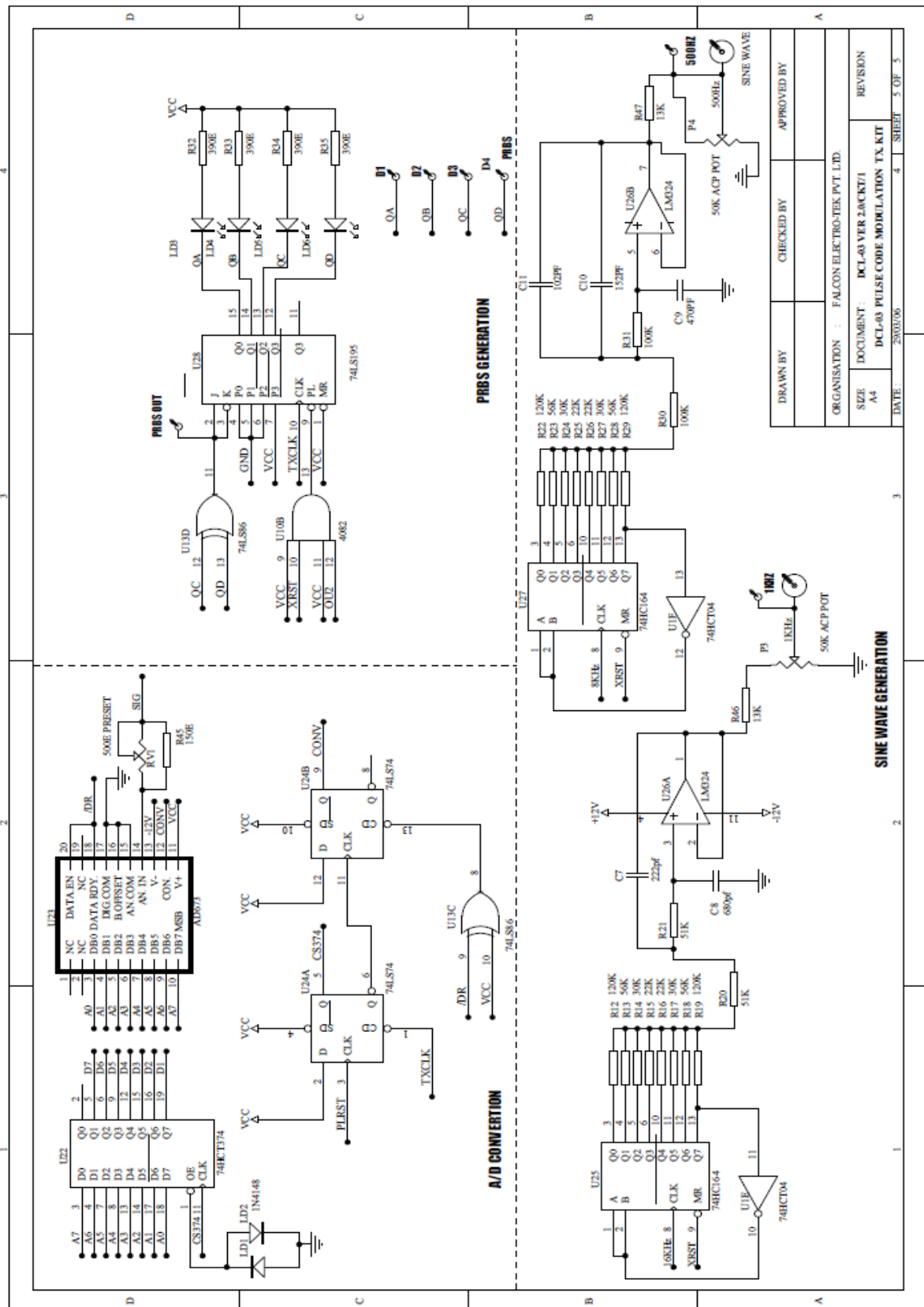
- Put switch **1** of **SF1 (DCL-03)** in Switch Fault section to ON position. This will introduce error in B4 bit of Parity Coded Data by complementing it at Error Code Generator logic.
- Put switch **2** of **SF1 (DCL-03)** in Switch Fault section to ON position. This will introduce error in B5 bit of Parity Coded Data by complementing it at Error Code Generator logic.
- Put switch **3** of **SF1 (DCL-03)** in Switch Fault section to ON position. This will introduce error in B6 bit of Parity Coded Data by complementing it at Error Code Generator logic.
- Put switch **4** of **SF1 (DCL-03)** in Switch Fault section to ON position. This will introduce error in B7 bit of Parity Coded Data by complementing it at Error Code Generator logic.
- Put switch **1** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove connection between EX-OR gate output & multiplexer output in Parity generator logic. Due to this there is no Parity indication in B1 for PARITY CODED DATA.
- Put switch **2** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove PRBS sequence from Transmitted Data. Synchronization will be only possible in case of direct connection of TXSYNC to RXSYNC. In any other case no synchronization is possible.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **3** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will open the connection of seventh bit of Error Detection and correction logic. Due to this fault inserted in B7 position is not recognized & not corrected in case of Hamming Parity mode.
- Put switch **4** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will remove CH0 filter capacitor connection from ground. Filter output will be distorted.

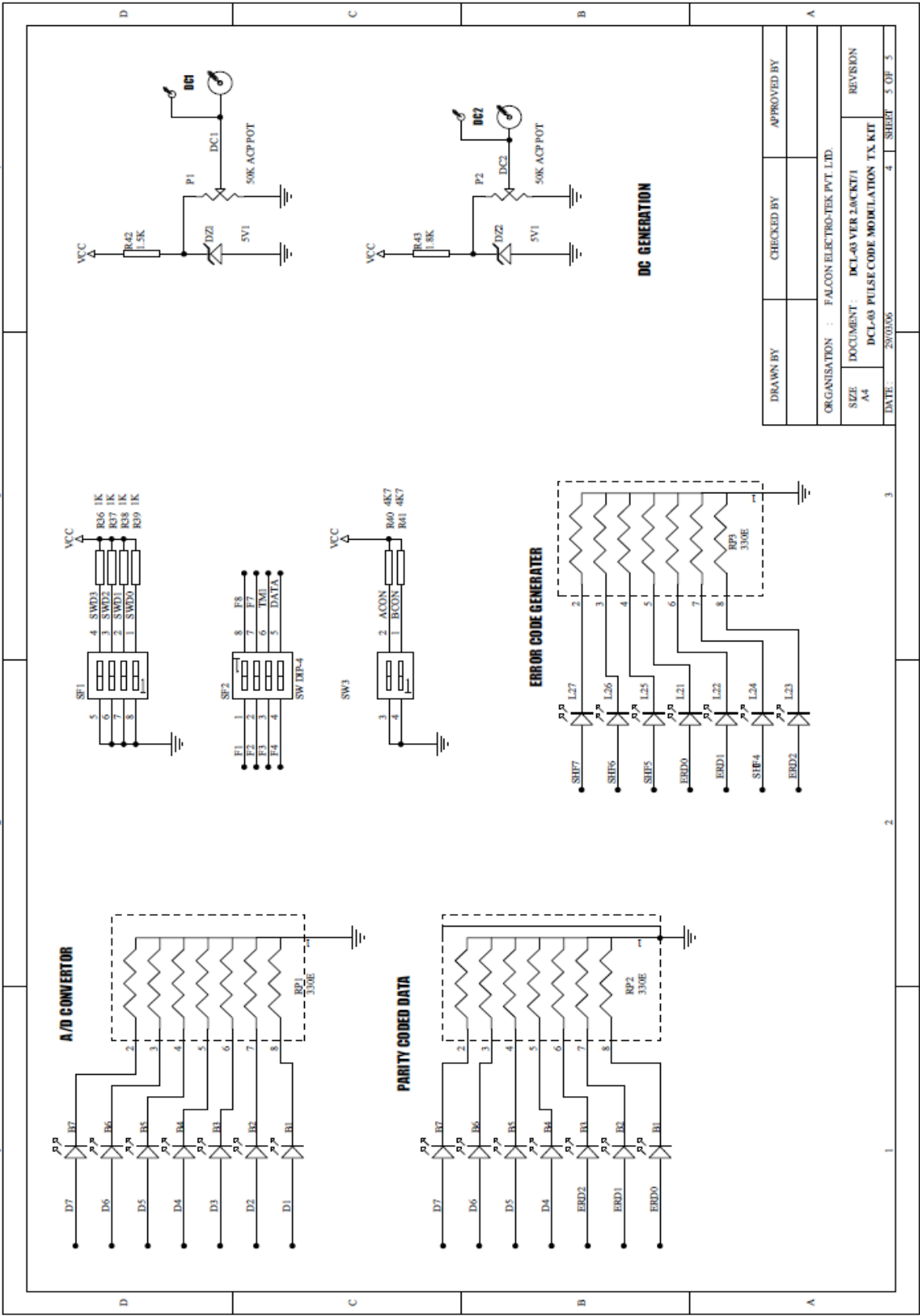
CIRCUIT DIAGRAM

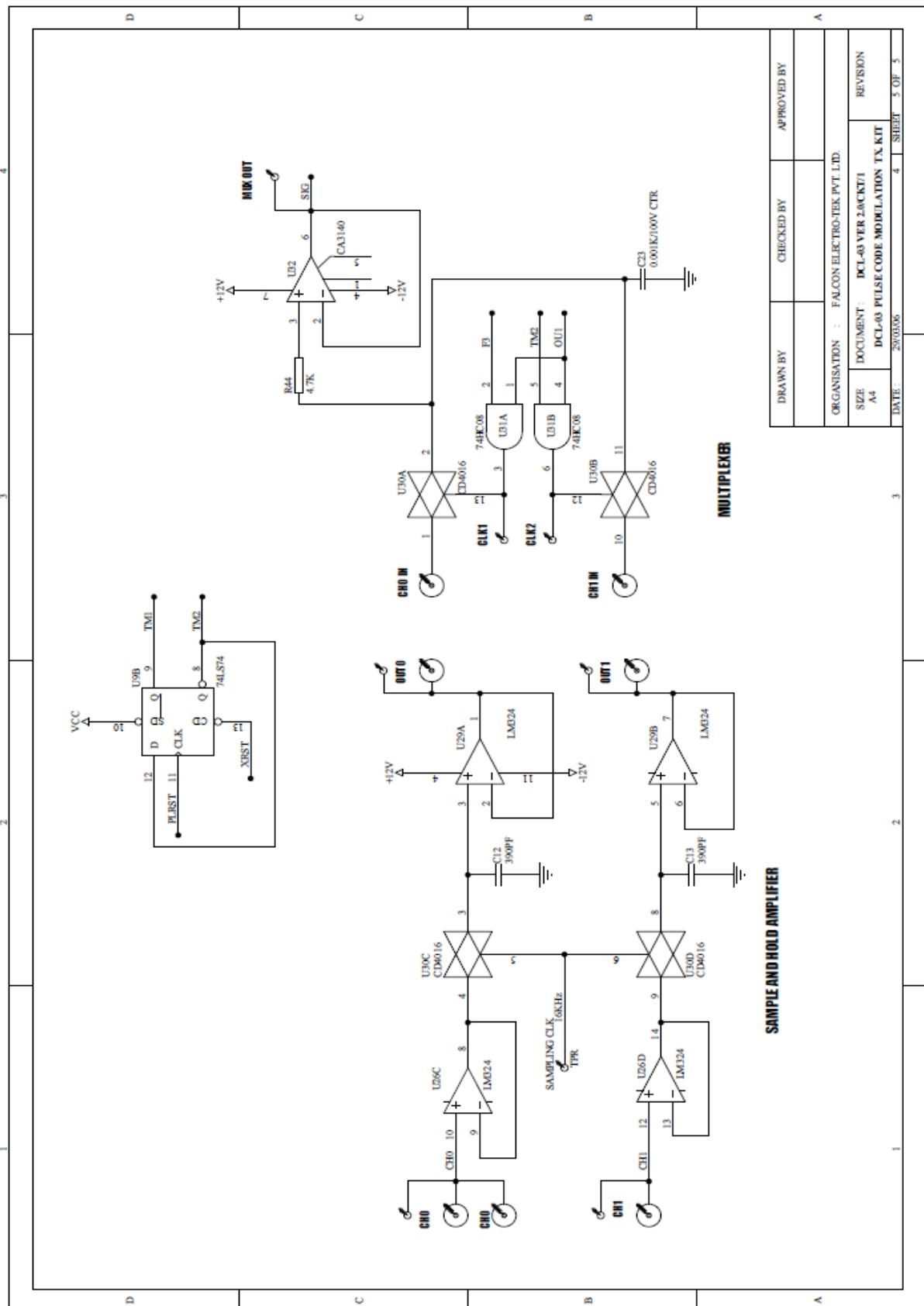
DCL-03

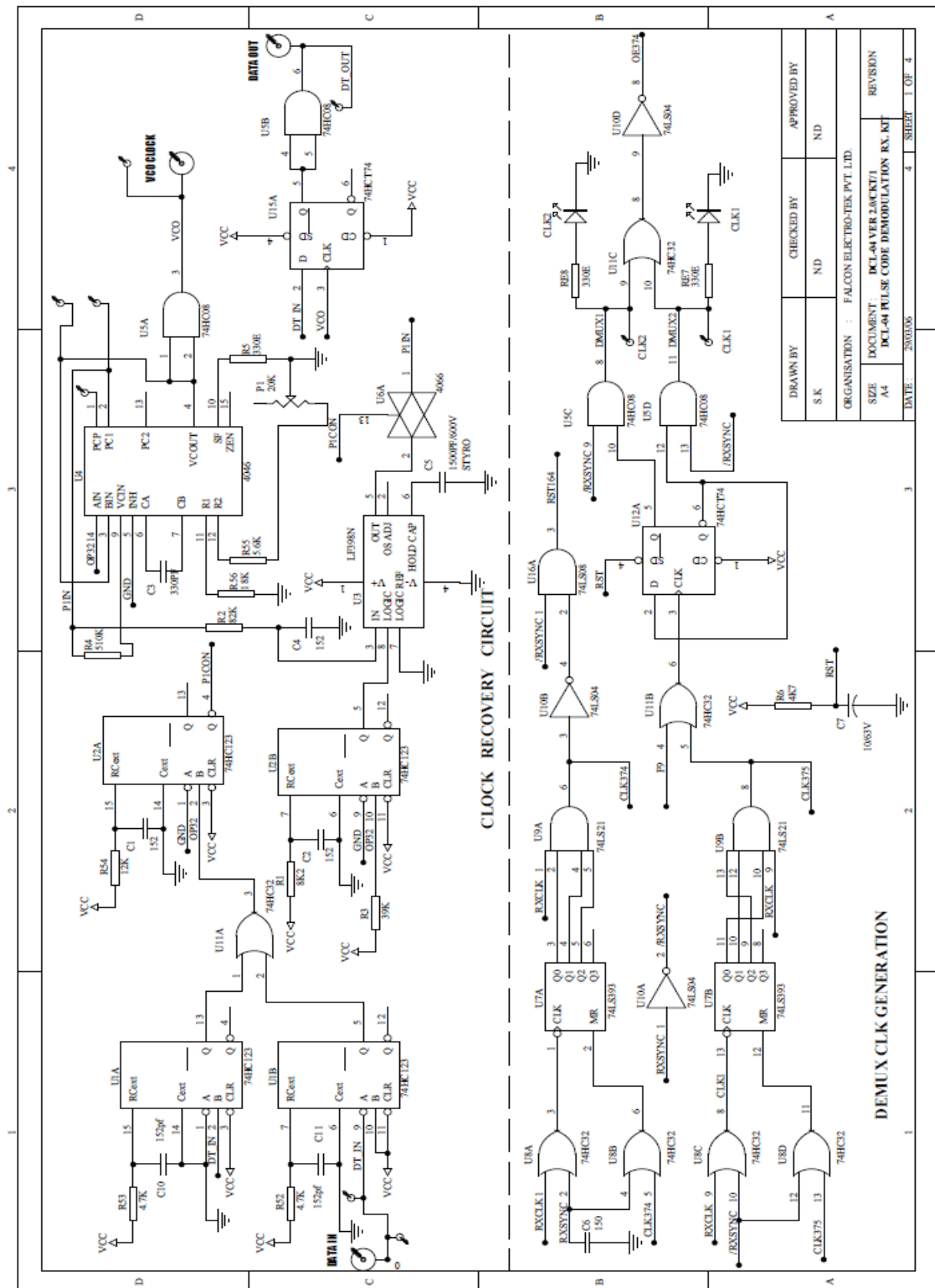


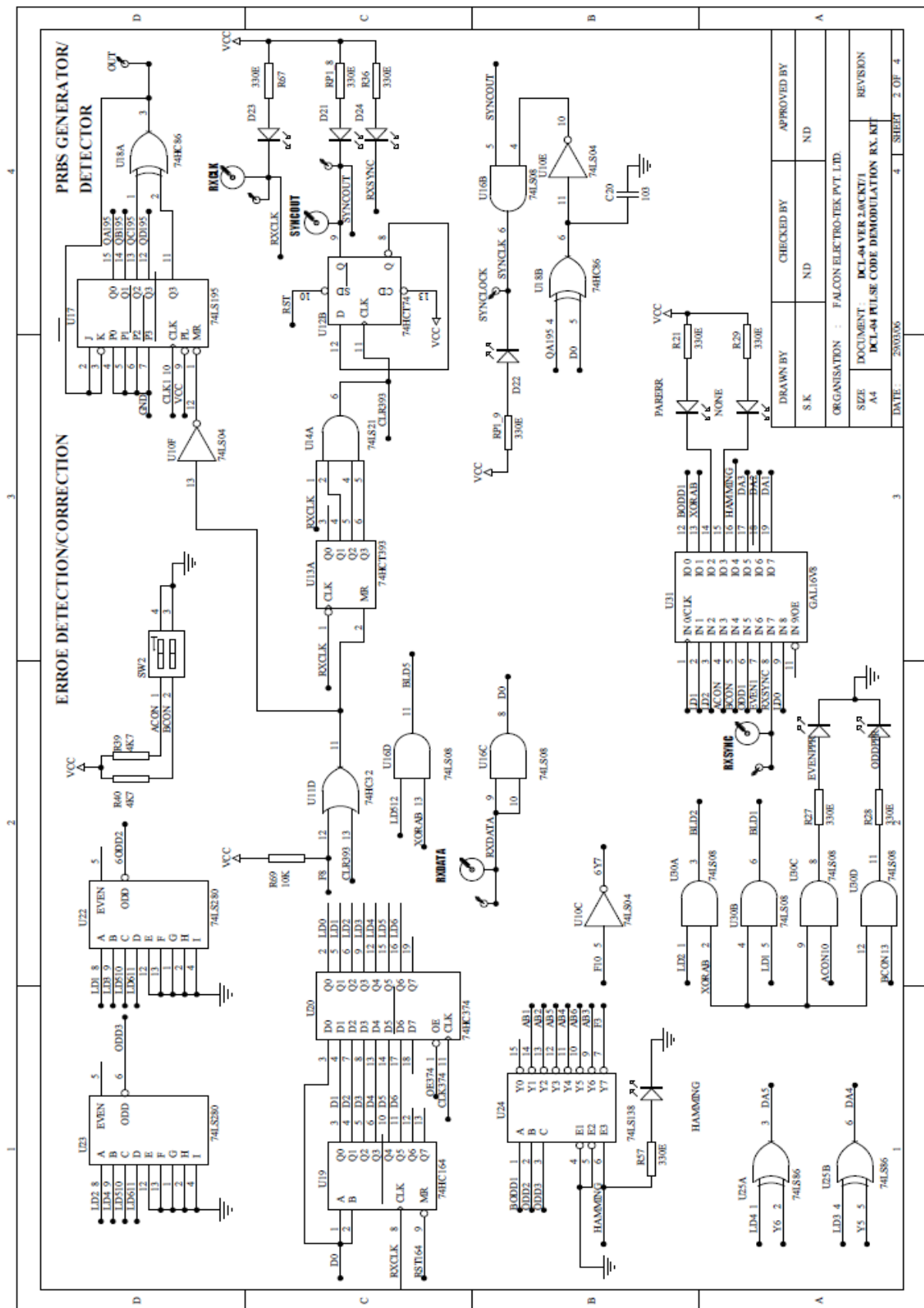


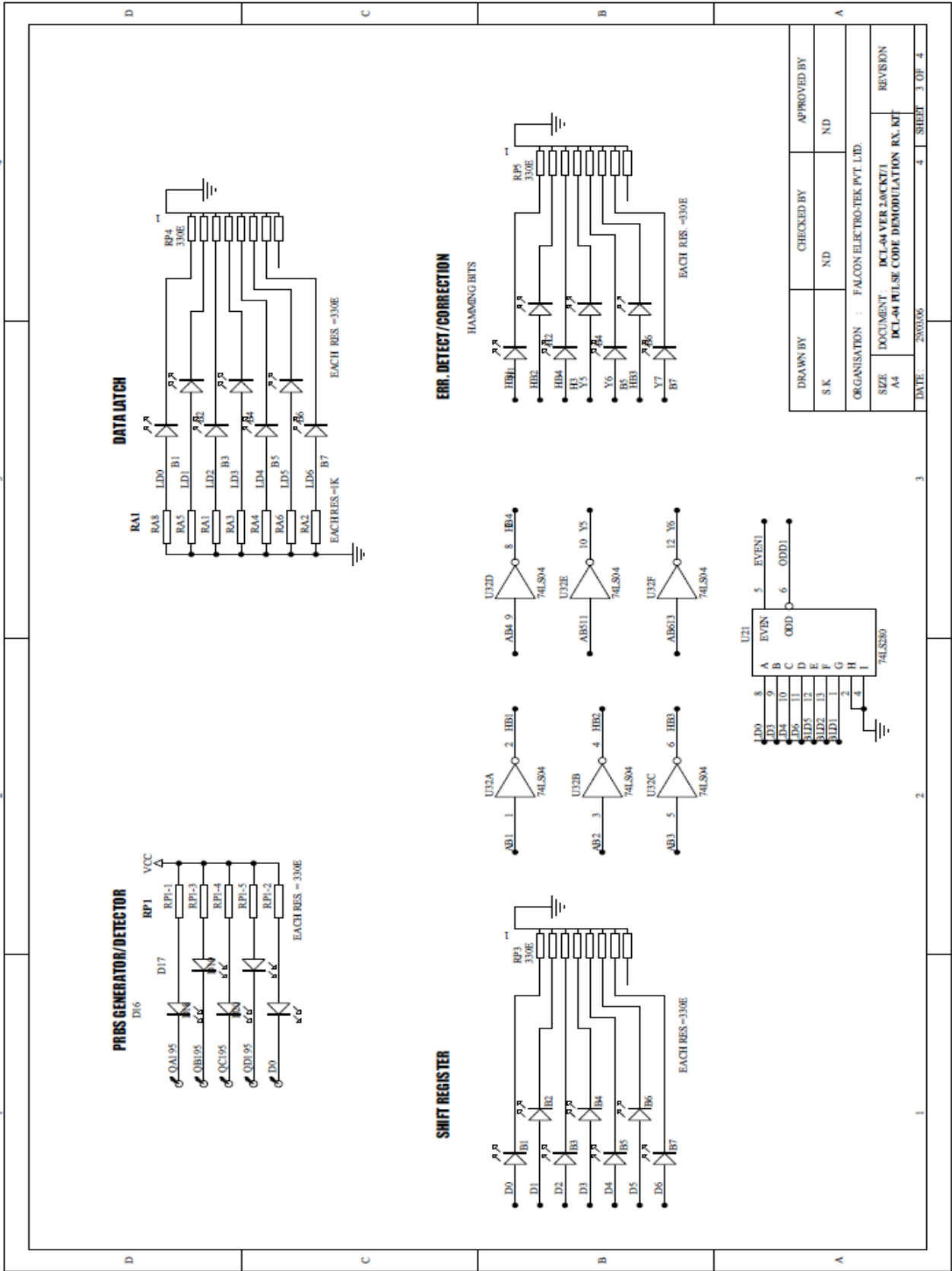




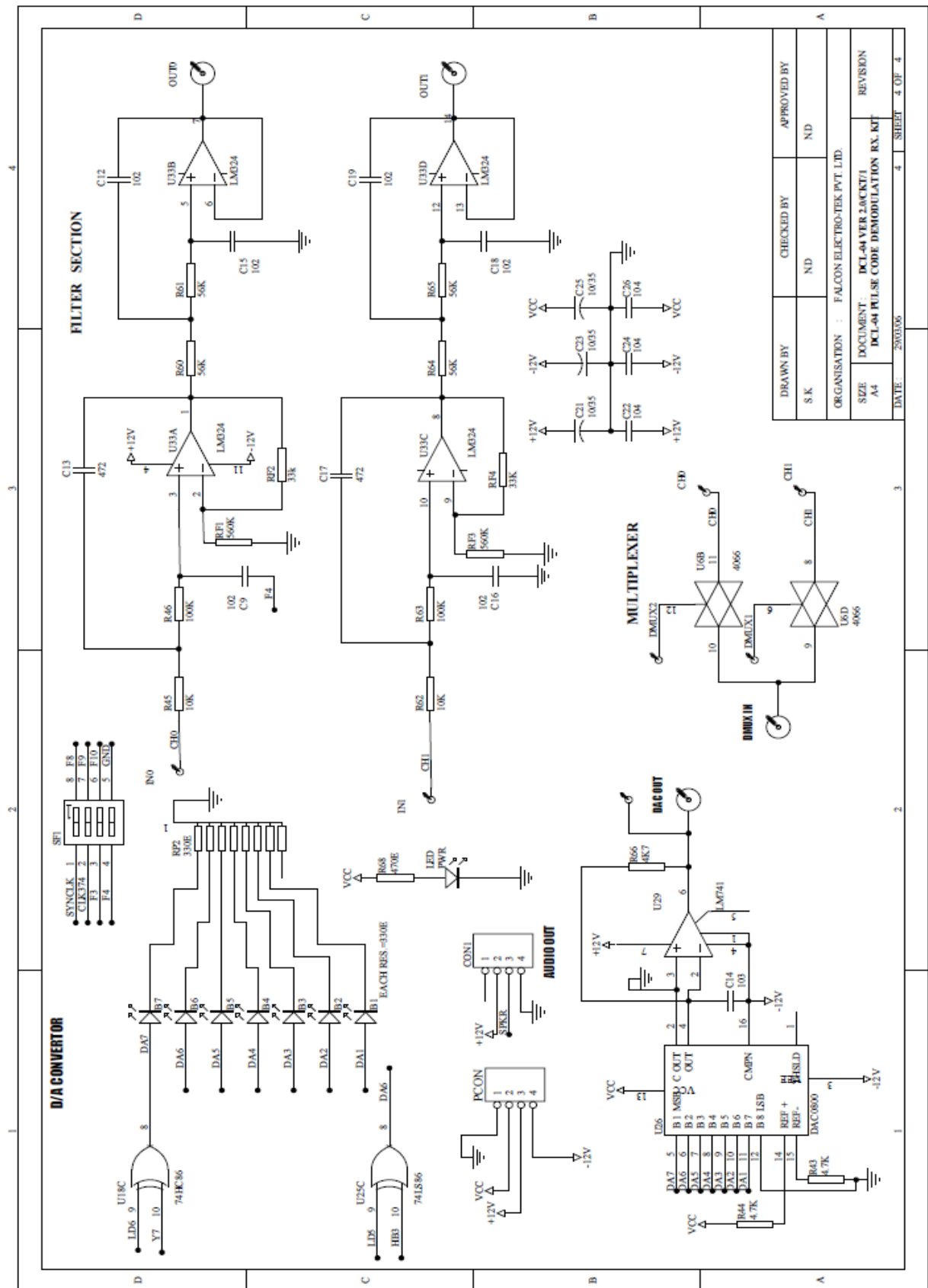


DCL-04

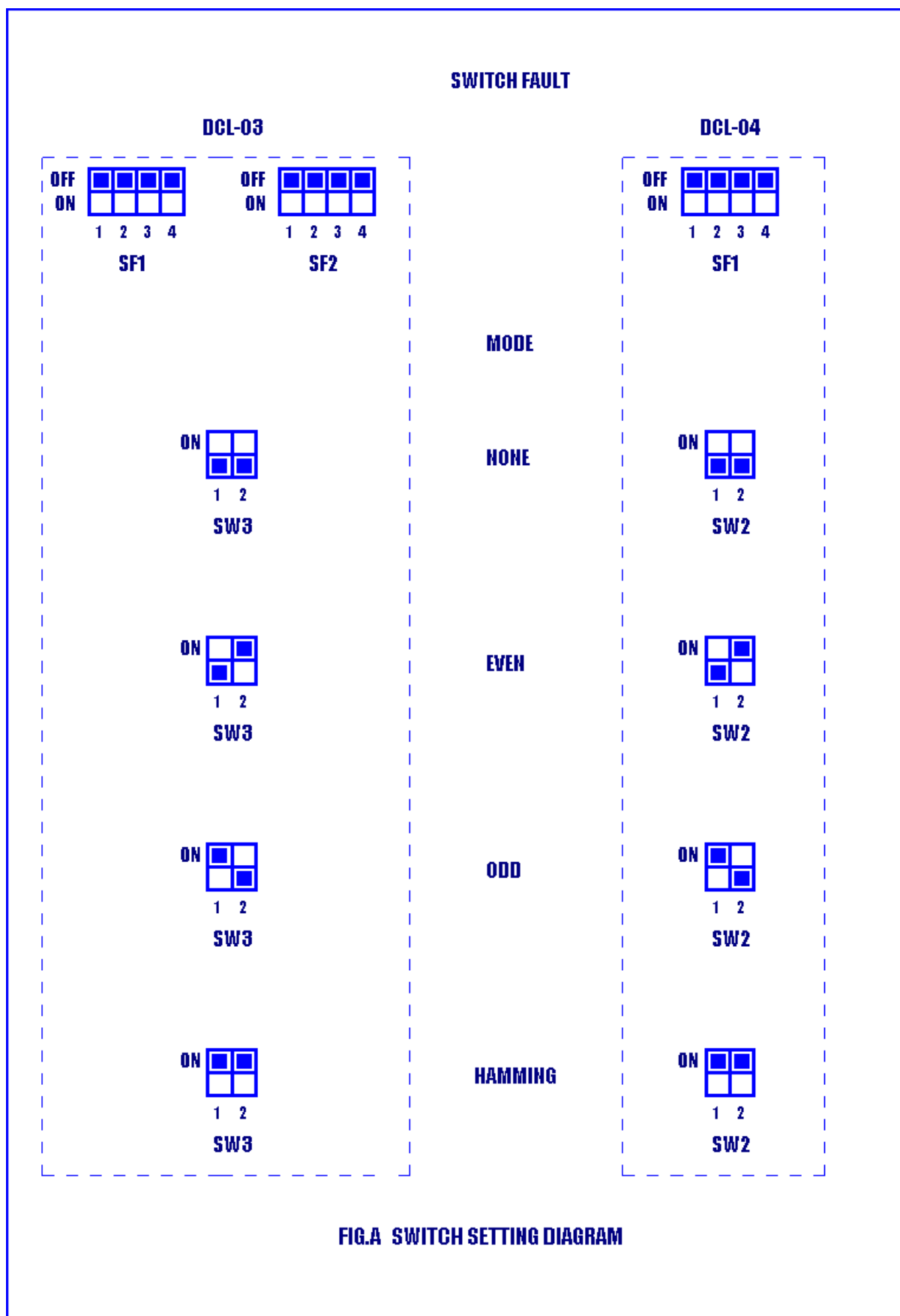




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S.K.	ND	ND
ORGANISATION	FALCON ELECTRO-TEK PVT. LTD.	
SIZE	DOCUMENT	REVISION
A4	DCL-04 VER 2.0 (KT/1)	
	DCL-04 PULSE CODE DEMODULATION RX KIT	
DATE	28/03/06	SHEET 3 OF 4



EXPERIMENT NO.1



EXPERIMENT NO: 1

NAME:

Sampling and Multiplexing Techniques and Reconstruction

OBJECTIVE:

To study 2-channel Time Division Multiplexing and Sampling of analog signal, and its pulse code modulation in none parity mode in the transmitter section and to study the demultiplexing and the reconstruction of the analog signal in the receiver section

THEORY:

The sine waves (analog signal) of frequency 500Hz and 1 KHz and DC Signal DC1 and DC2 whose amplitude can be varied accordingly are generated onboard on DCL-03. These signals are fed to the input of the Sampling logic CH0 & CH1 and their samples are multiplexed by interleaving them properly in their assigned time slots.

The crystal oscillator generates a clock of 6.4MHz from which all the transmitter data and timing signals are derived. For fast mode operation the transmitter clock is 240 KHz, and sampling clock is 16 KHz. For slow mode operation depending on jumper position the transmitter clock is 1.23Hz or 0.6Hz and sampling clock is 0.088Hz or 0.044Hz i.e. the sampling rate per channel is 11 or 22 seconds and serial data transmission rate is 813 milliseconds or 1.6 seconds.

The multiplexed data is Pulse Code Modulated before transmission. At the receiver after the Pulse Code Demodulation, the recovered multiplexed data is sent to De-multiplexing logic. The two demultiplexed samples are fed to reconstruction unit. Which consist of 4th order Low Pass Butterworth Filter, where frequency components are filtered out to recover the original base band signal at the receiver output CH0 and CH1

EQUIPMENTS:

- DCL-03 & DCL-04 Kit.
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

1. Refer to the Block Diagram (Fig. 1.1) & Carry out the following connections.
2. Connect power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
3. Connect sine wave of frequency **500Hz** and **1 KHz** to the input **CH0** and **CH1** of the sample and hold logic.

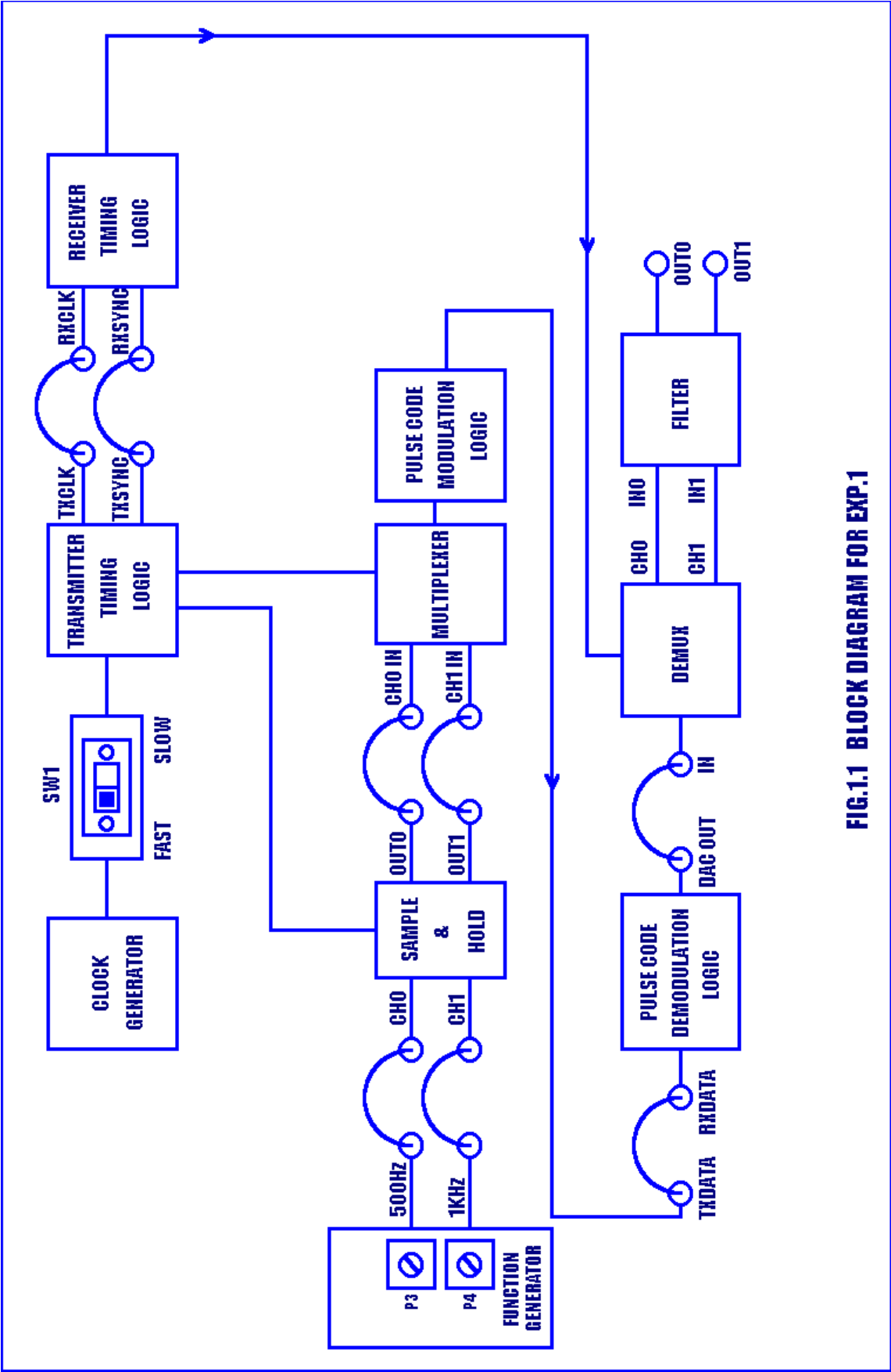


FIG.1.1 BLOCK DIAGRAM FOR EXP.1

4. Connect **OUT 0** to **CH0 IN** & **OUT 1** to **CH1 IN**.
5. Set the speed selection switch **SW1** to **FAST** mode.
6. Select parity selection switch to **NONE** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
7. Connect **TXDATA**, **TXCLK** and **TXSYNC** of the transmitter section **DCL-03** to the corresponding **RXDATA**, **RXCLK**, and **RXSYNC** of the receiver section **DCL-04**.
8. Connect posts **DAC OUT** to **IN** post of de-multiplexer section on **DCL-04**.
9. Ensure that **FAULT SWITCH SF1** as shown in switch setting diagram (Fig. A) introduces no fault.
10. Take the observations as mentioned below.
11. Repeat the above experiment with **DC** Signal at the inputs of the Channel **CH 0** and **CH 1**.
12. Connect ground points of both the kits with the help of connecting chord provided during all the experiments.

OBSERVATION:

Observe the following signal on oscilloscope and plot it on the paper.

ON KIT **DCL-03** (Fig. 1.2) & (Fig. 1.3)

- Input signal **CH 0** and **CH 1**.
- Sample and Hold output **OUT 0** and **OUT1**
- Multiplexer clock **CLK 1** and **CLK 2**
- Multiplexed data **MUX OUT**.
- PCM Data **TX DATA**, **TXCLK**, **TXSYNC**

ON KIT **DCL-04** (Fig. 1.4) & (Fig. 1.5)

- **RXCLK**, **RXSYNC**, **RXDATA**
- **DAC OUT**
- Demultiplexer clock **CLK 1** and **CLK 2**
- Demultiplexed Data **CH 0** and **CH 1**
- Received signal **OUT 0** and **OUT 1**

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

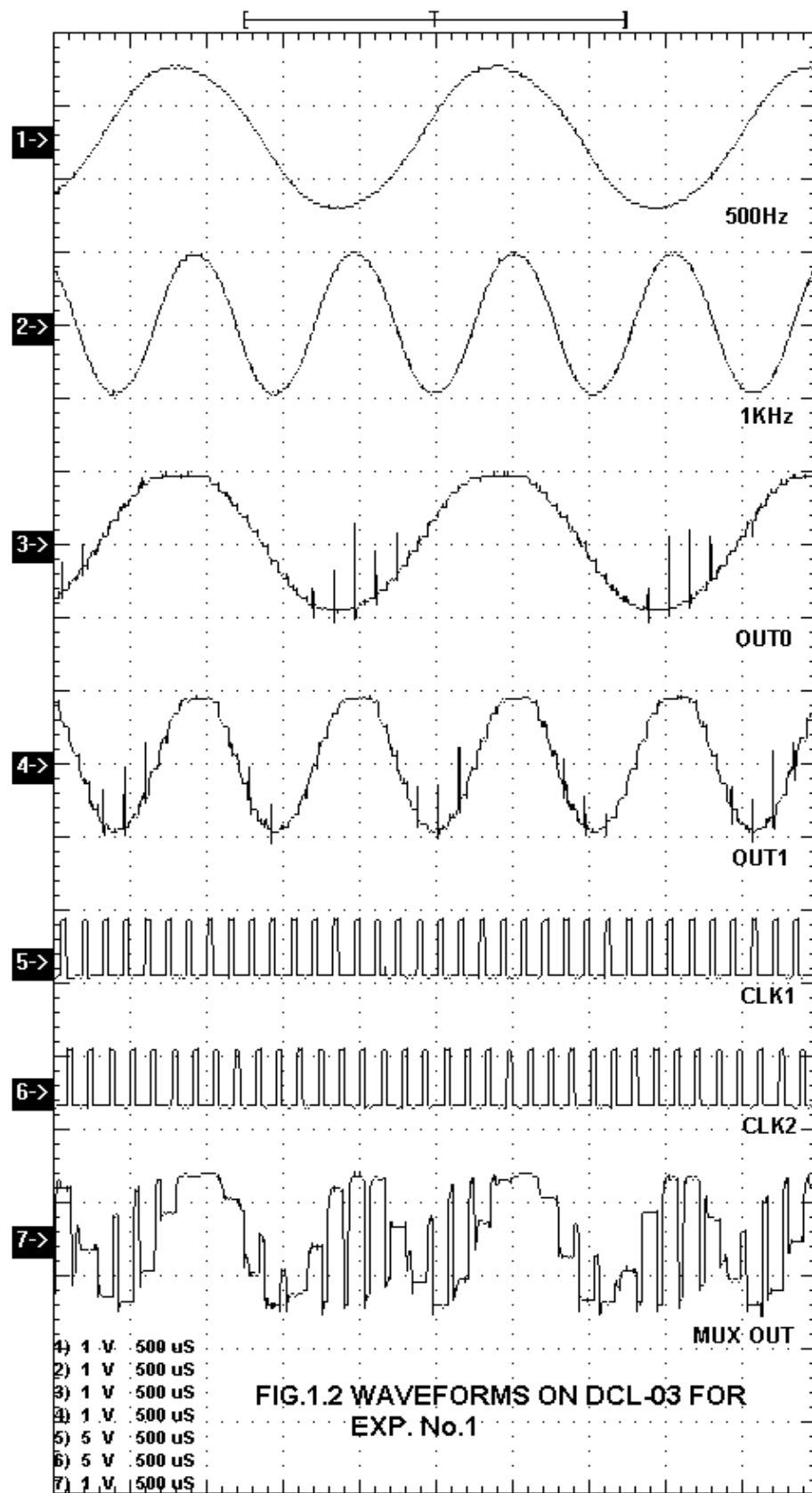
- Put switch **2** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove PRBS sequence from Transmitted Data. Synchronization will be only possible in case of direct connection of TXSYNC to RXSYNC. In any other case no synchronization is possible.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will Increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.

- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **4** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will remove CH0 filter capacitor connection from ground. Filter output will be distorted.

CONCLUSION:

We conclude that at the transmitter side sampling for 500Hz and 1 KHz signals is done by using 16 KHz sampling clock, hereby satisfying the Nyquist criterion. Similarly the multiplexed output observed in the oscilloscope shows the proper alignment of samples in their respective time slots.

At the receiver side the 4th order low pass butter-worth filter is used as reconstruction unit, which reproduce the signals (sine wave and DC signal levels) same as that of the transmitter side. It is observed in this case, that the reconstructed sine wave has good linearity.



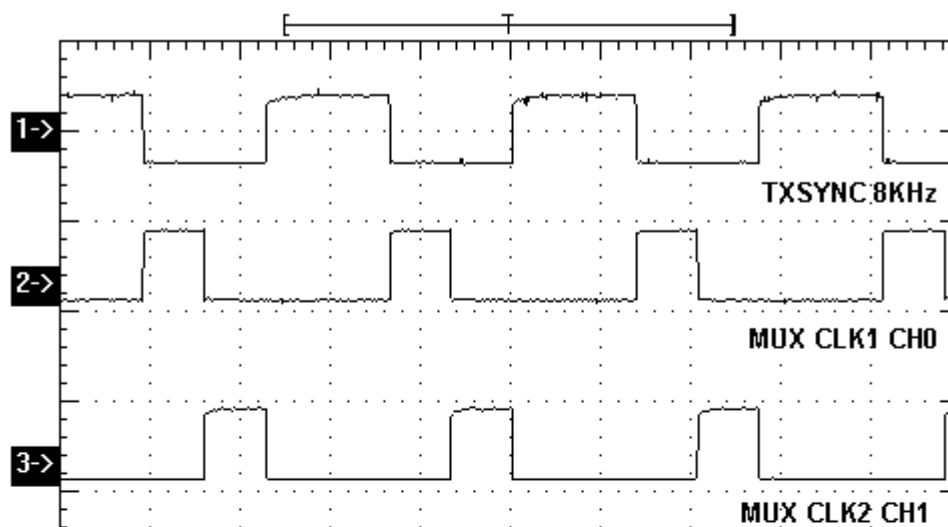


FIG.1.3 WAVEFORMS FOR TX TIMING

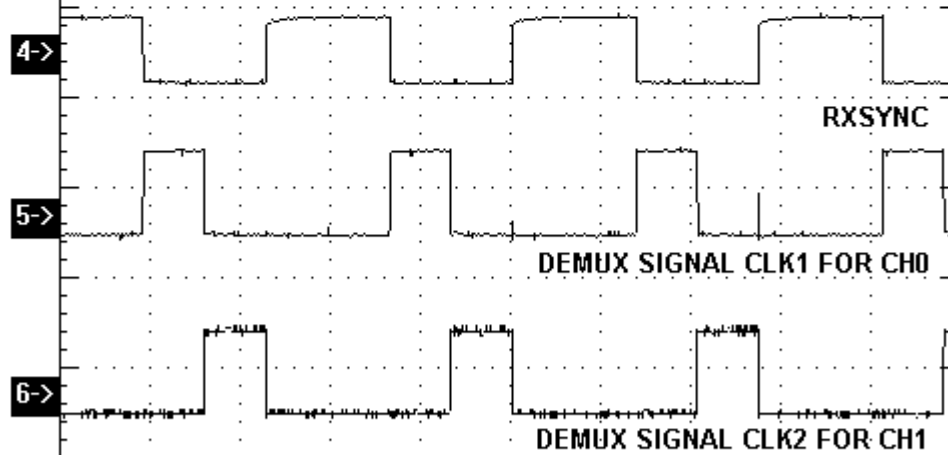
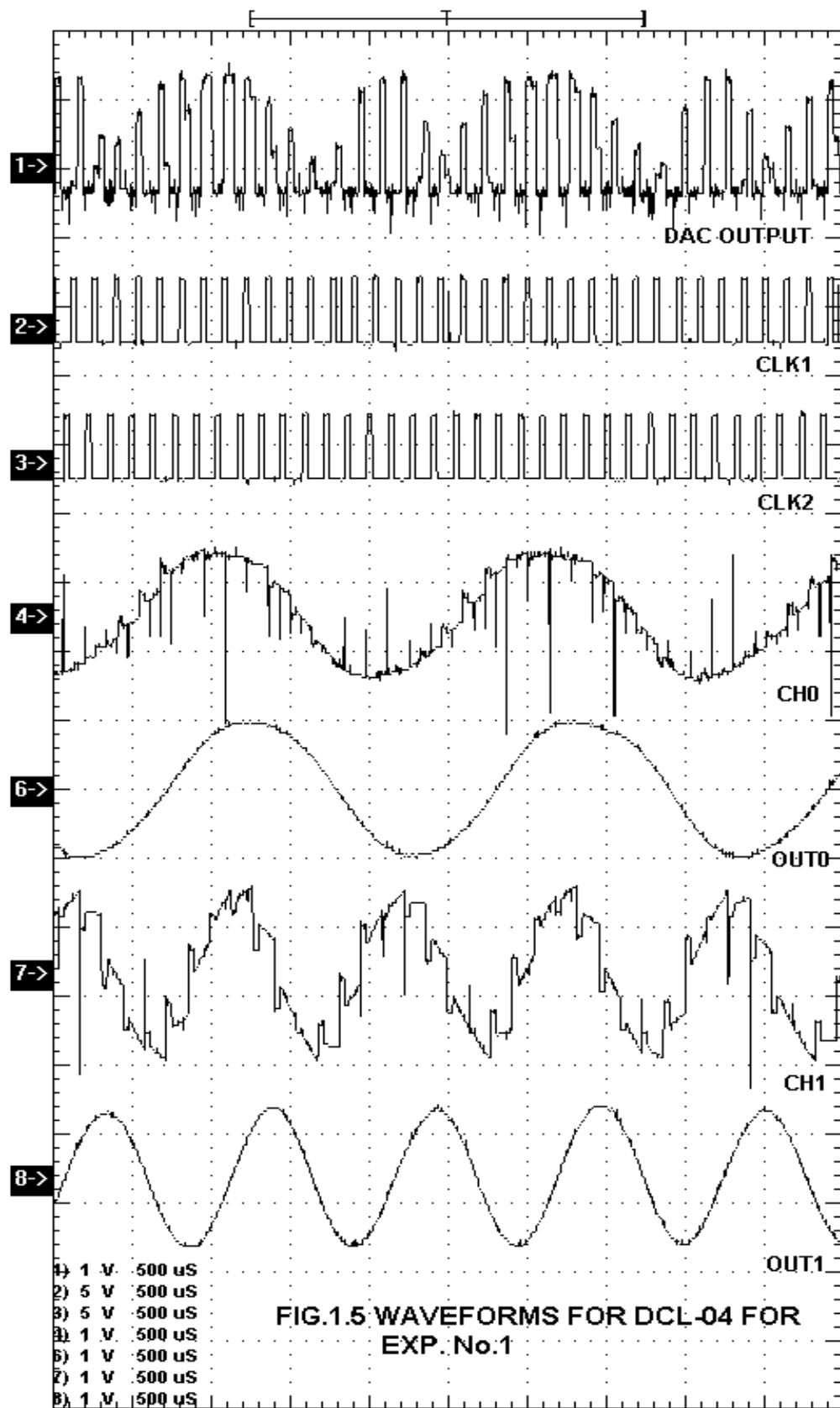
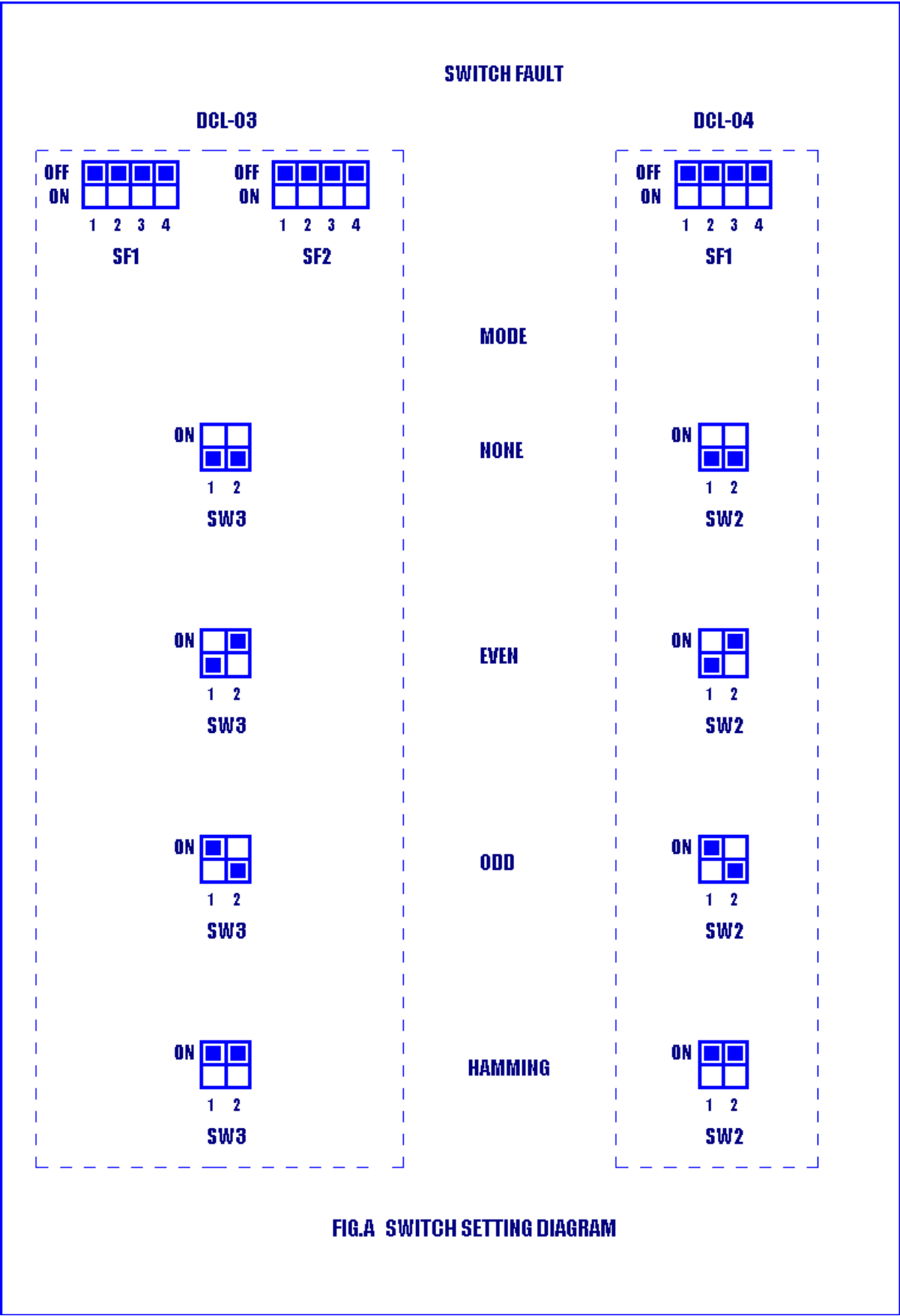


FIG.1.4 WAVEFORMS FOR RX TIMING

- 1) 5 V 50 μ S
- 2) 5 V 50 μ S
- 3) 5 V 50 μ S
- 4) 5 V 50 μ S
- 5) 5 V 50 μ S
- 6) 5 V 50 μ S



EXPERIMENT NO.2



EXPERIMENT NO: 2

NAME:

Principle of A/D Converter and D/A Converter

OBJECTIVE:

To Study Principle of Analog to Digital and Digital to Analog Conversion

THEORY:

The Analog to Digital Converter converts the analog samples to digital bits. The analog to digital converter IC AD673 forms the heart of this logic; this device performs both the quantising and encoding operations.

The level of the input signal can be varied from 0 volt to 4.96 volts. The level is dividing to 128 uniform steps. Each step corresponds to $4.96\text{V} / 128 = 40\text{ mV}$.

The quantising level is considered to be midway of the steps. If the signal level falls below the quantising level, then the signal is rounded off to the lower level. If the signal level falls above the quantising level, then the signal is rounded off to the upper level. This type of quantising is called "Uniform quantising, where the step levels are uniform. Then corresponding to the level chosen, code words are assigned to the samples. The code words vary from 0000000 to 1111111 for the 0 to 4.96 V.

The timing of the A/D conversion is controlled by the convert pulse, which is generated in the transmitter timing logics. As soon as the data is ready, the AD673 gives a data ready (DR Active low) output. The data latch is used for latching the valid outputs of the A/D converter. The timing and latching interval of the data latch is controlled by the latch enable signal LE374*. Each sample is coded to a 7 bit data by the AD673.

Thus the Analog to Digital converter assigns the code words for all the samples.

The Digital to Analog Converter Logic converts the 7 bit coded sequence obtained at the output of the detect/correct logic to analog samples. This logic is built around the Digital to Analog Converter IC, DAC0800. It reproduces the multiplexed sample levels at the receiver.

EQUIPMENT:

- DCL-03 & DCL-04 Kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep the Switch Faults In Off Position.

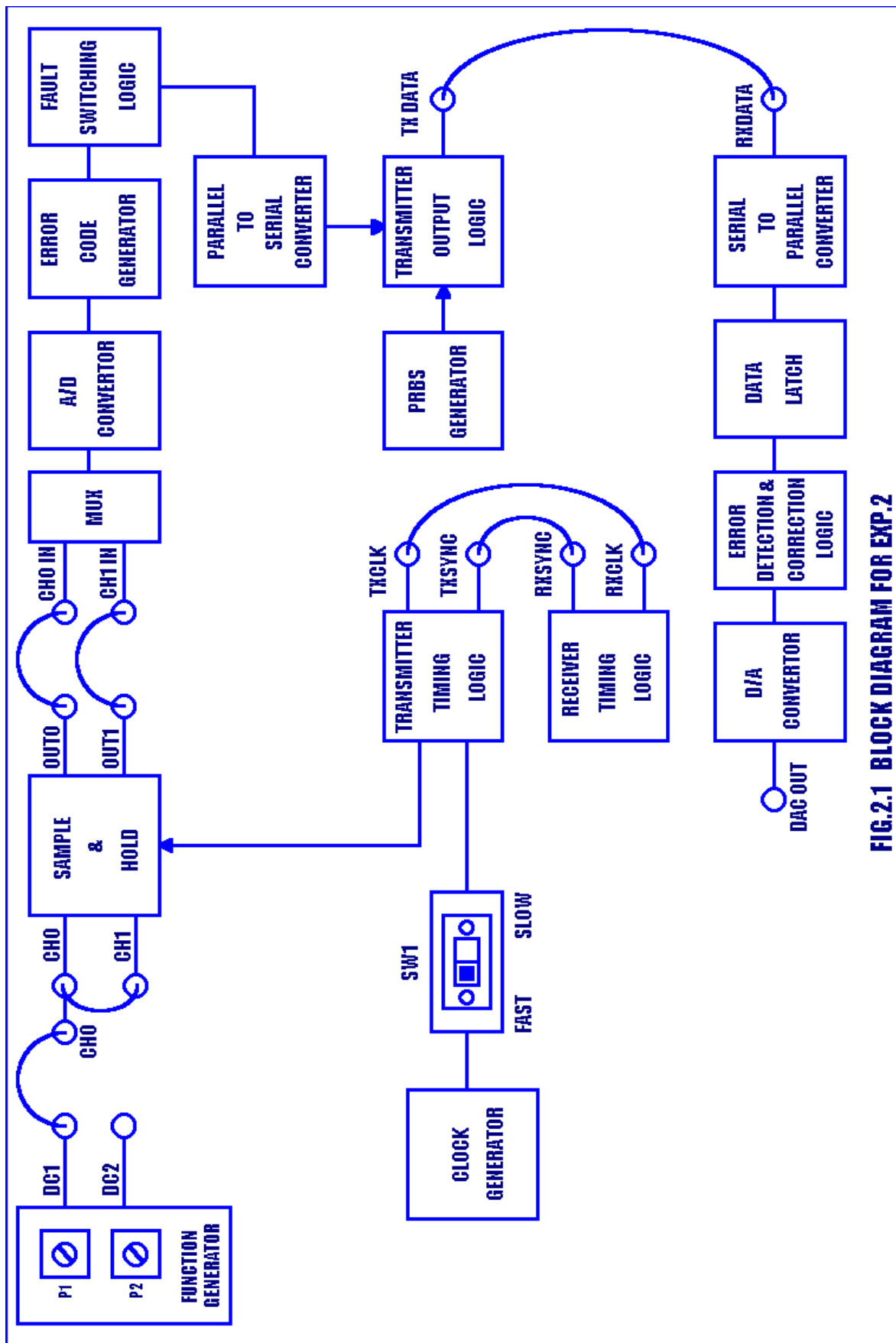


FIG.2.1 BLOCK DIAGRAM FOR EXP.2

PROCEDURE:

1. Refer to the Block Diagram (Fig. 2.1) & Carry out the following connections and switch settings.
2. Connect power supply with proper polarity to the kits **DCL-03** and **DCL-04** and switch on.
3. Connect DC input signal **DC 1** to the input **CH 0** and **CH 1** of the Sample and Hold logic.
4. Connect **OUT 0** to **CH0 IN** & **OUT 1** to **CH1 IN**.
5. Set the speed selection switch **SW1** to **FAST** mode.
6. Select parity selection switch to **NONE** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
7. Connect **TXDATA**, **TXCLK** and **TXSYNC** of the transmitter section **DCL-03** to the corresponding **RXDATA**, **RXCLK**, and **RXSYNC** of the receiver section **DCL-04**.
8. Ensure that **FAULT SWITCH SF1** as shown in switch setting diagram (Fig. A) introduces no fault.
9. Vary the amplitude of input DC signal from 0V to 4.96V.
10. Take the observations as mentioned below.
11. Repeat the experiment in the **SLOW** mode.

OBSERVATIONS:

Observe the following signal for each setting of input DC signal and tabulate at least ten reading in your record book.

ON KIT **DCL-03**

- Input Signal to ADC (On CRO or Digital Multimeter) **MUX OUT**
- ADC Output (On LED's) **B1** to **B7**

ON KIT **DCL-04**

- DAC Input (On LEDs) **B1** to **B7**
- DAC Output Signal (On CRO or Digital Multimeter) **DAC OUT**

OBSERVATION TABLE: (Fig. 2.2)

ADC Input (Volts)	ADC Output	DAC Input	DAC Output (Volts)
0 volts	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 Volt
4.96 volts	1 1 1 1 1 1 1	1 1 1 1 1 1 1	4.96 Volts

Draw a graph between the A/D input voltage and the A/D output data based on the tabulations. From the staircase waveform derived, determine the quantising step, which is about 40mV for AD673.

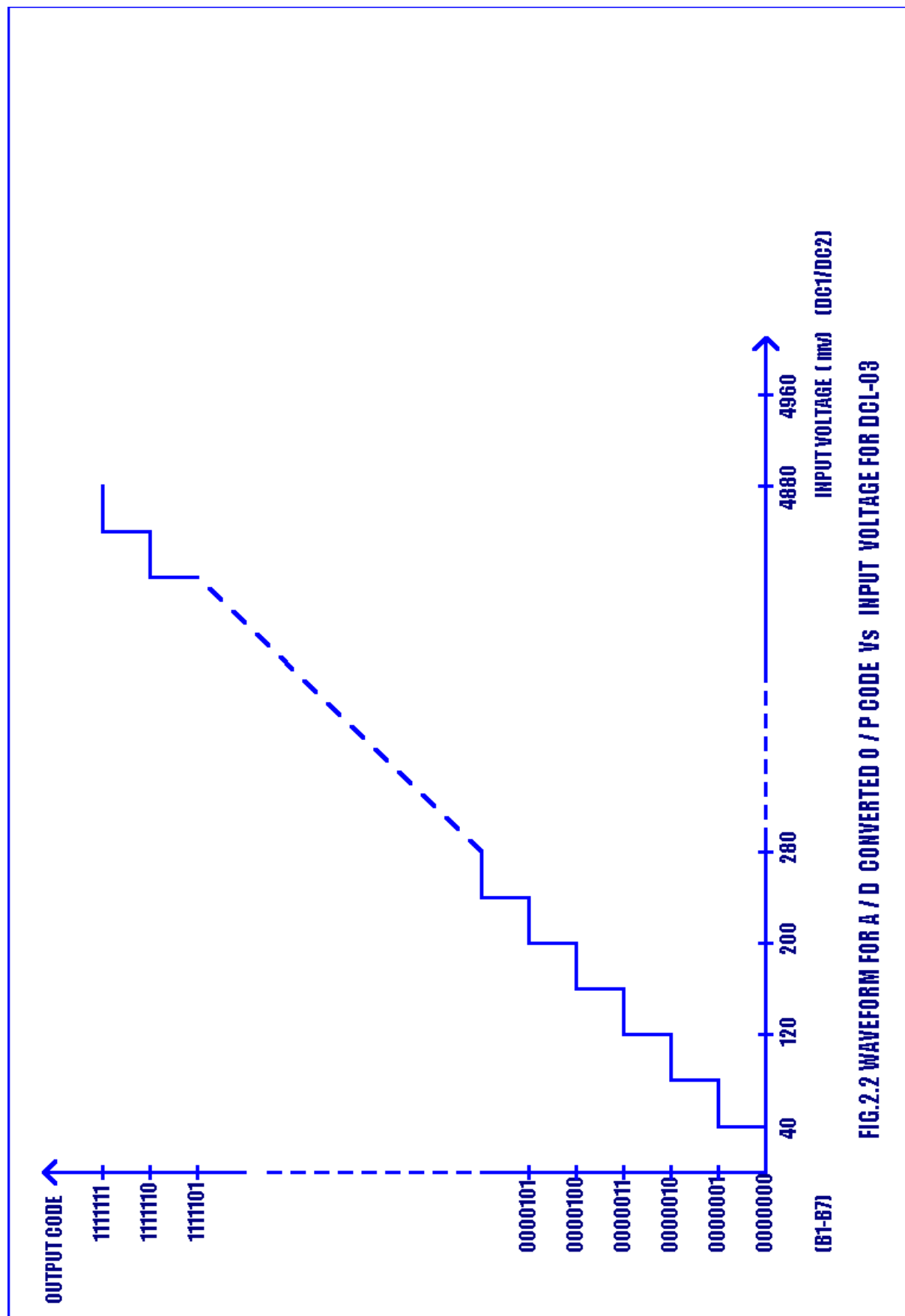
It has been observed that the amplitude level at the output of the receiver is same as the input level of the transmitter.

Note: Always a single bit error (LSB) is common in the analog to Digital Conversion applications.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.



EXPERIMENT NO.3

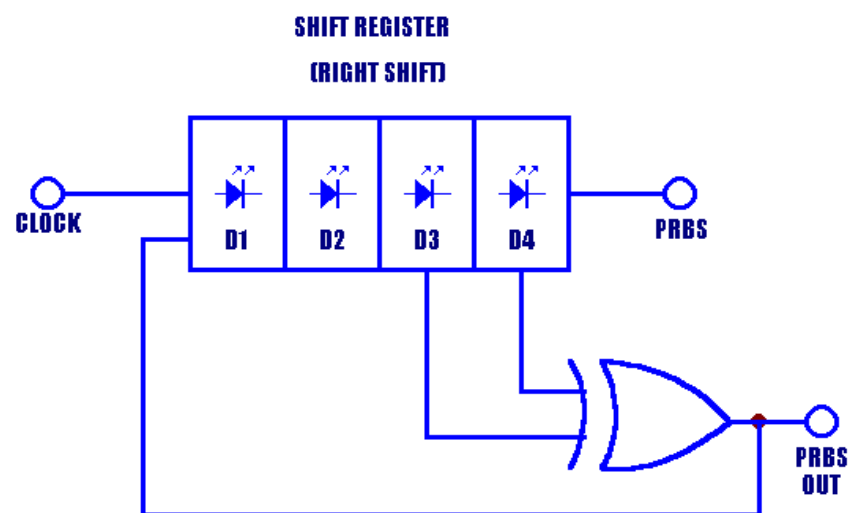
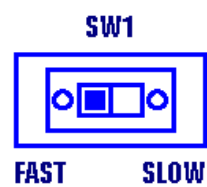


FIG.3.1 BLOCK DIAGRAM FOR PRBS GENERATOR



EXPERIMENT NO:3

NAME:

Pseudo Random Sequence

OBJECTIVE:

Study of Pseudo Random Sequences

THEORY:

For ensuring proper Frame Synchronization between the transmitter and the receiver, a 14-bit Pseudo Random sequence is transmitted along with the data. A 4-bit Shift Register with an EX-OR gate at the feedback generates the Pseudo Random sequence, which is a unique pattern that has minimum correlation with the data. The clock frequency for the Pseudo Random Sequence Generator is the same as the frequency of the transmitted serial data. Note that PRBS is generated and transmitted only when the sync remains high. When the sync is low, no PRBS is generated and at that time, the serial data is transmitted over the line.

The PRBS sequence generated on the kit DCL-03 is of the following format:

1 0 0 0 1 0 0 1 1 0 1 0 1 1

EQUIPMENT:

- DCL-03 & DCL-04 Kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep the Switch Faults In Off Position

PROCEDURE:

PART A

1. Connect power supply in proper polarity to the kit **DCL-03** and switch it on.
2. Set the speed selection switch **SW1** to **FAST** mode.

OBSERVATION:

Using an oscilloscope, see the waveforms at the test points provided for the observations of Pseudo Random sequence. While observing the Pseudo Random Sequence, use two channel oscilloscope in one channel observe the Synchronization signal **TXSYNC** and in another channel observe the Pseudo Random Sequence at **D4**. (Fig. 3.2)

Whenever the sync pulse remains high, the pseudo random code is generated. Identify the levels with respect to the clock and write down the levels to verify with the PRBS truth table.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **2** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove PRBS sequence from Transmitted Data. Synchronization will be only possible in case of direct connection of TXSYNC to RXSYNC. In any other case no synchronization is possible.

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

PART B

- Connect power supply in proper polarity to the kit **DCL-03** and switch it on.
- Set the speed selection switch **SW1** to **SLOW** mode.

OBSERVATION:

Observe the flow of the Pseudo Random Sequence in LEDs **D1, D2, D3, and D4**

The flow in LEDs is observed to follow the sequence. Here also the student is advised to verify the truth table. The Pseudo Random Sequence is observed for 14 clock states and the LED remains OFF for another 14 clock states. Whenever the sync remains high (LED off), we can observe the shifting operations visually in LEDs (Note: LED display for PRBS sequence is complimentary.)

TRUTH TABLE

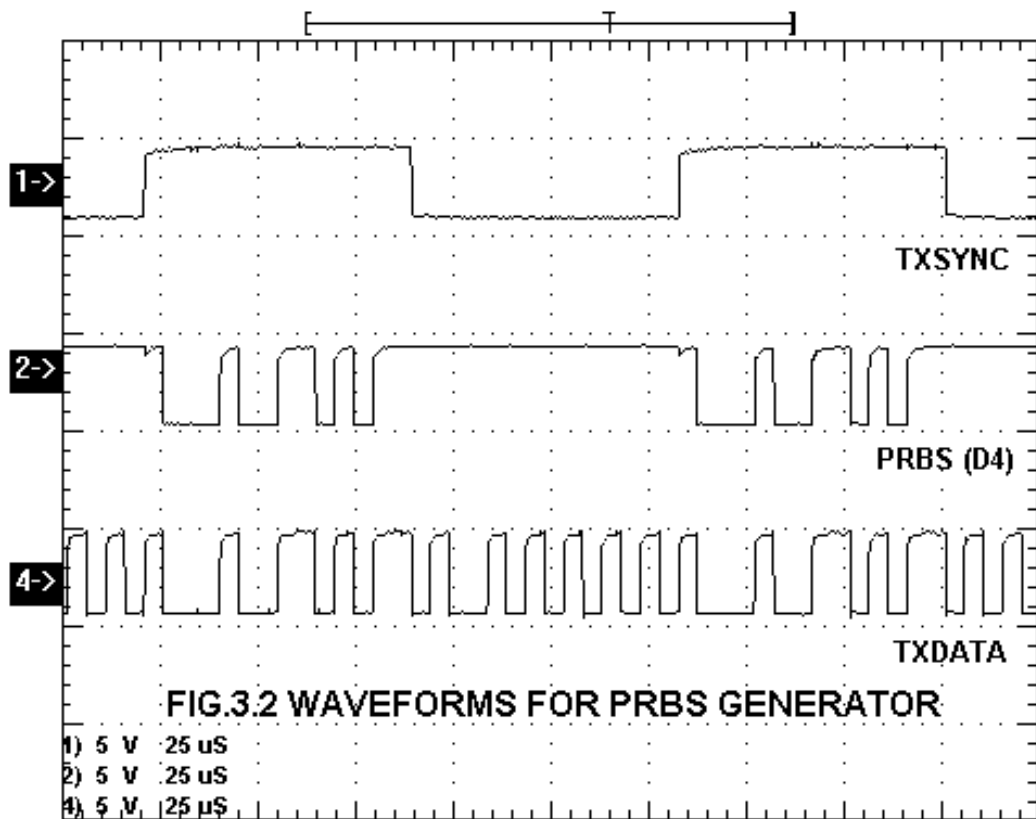
	<u>Theoretical</u>				<u>Practical</u>			
Clock	D1	D2	D3	D4	D1	D2	D3	D4
1	1	0	0	0	0	1	1	1
2	0	1	0	0	1	0	1	1
3	0	0	1	0	1	1	0	1
4	1	0	0	1	0	1	1	0
5	1	1	0	0	0	0	1	1
6	0	1	1	0	1	0	0	1
7	1	0	1	1	0	1	0	0
8	0	1	0	1	1	0	1	0
9	1	0	1	0	0	1	0	1
10	1	1	0	1	0	0	1	0

11	1	1	1	0	0	0	0	1
12	1	1	1	1	0	0	0	0
13	0	1	1	1	1	0	0	0
14	0	0	1	1	1	1	0	0
15	0	0	0	1	1	1	1	0

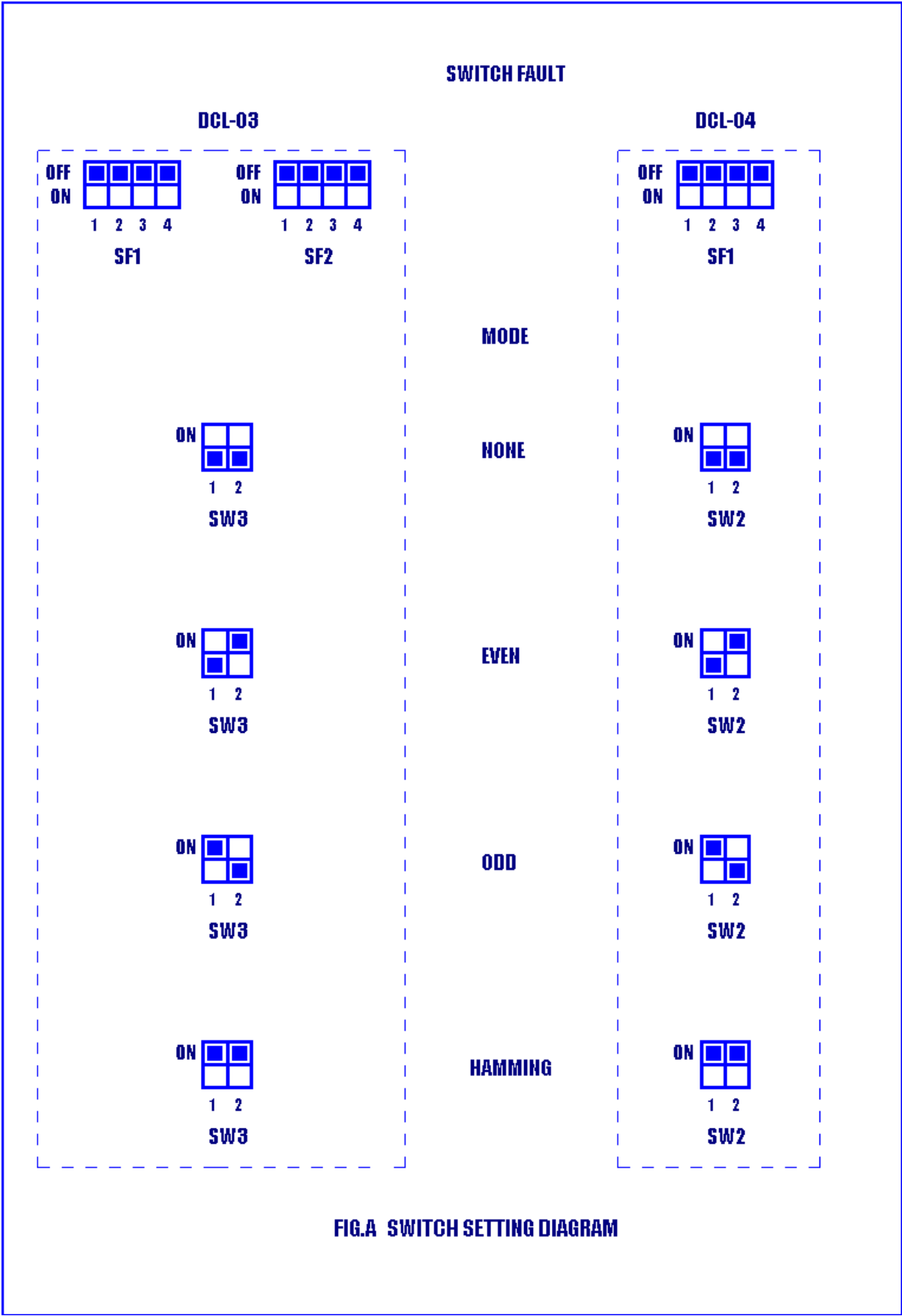
SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **2** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove PRBS sequence from Transmitted Data. Synchronization will be only possible in case of direct connection of TXSYNC to RXSYNC. In any other case no synchronization is possible.



EXPERIMENT NO.4



EXPERIMENT NO: 4

NAME:

Error Control Coding Techniques

OBJECTIVE:

Study of Error Check Code Logic

- Odd Parity Coding
- Even Parity Coding
- Hamming Coding

THEORY

Odd Parity Mode

ODD Parity Generator generates the coded sequence, which consists of 6 bits of data plus one ODD parity Bit for transmission. The least significant bit (B1) of the data word from the ADC is replaced with the parity bit. The coded sequence will be:

Odd Parity bit, B2, B3, B4, B5, B6, B7 (MSB)

Odd Parity Checker checks the number of Ones in the data stream and determines if a single bit error had occurred during transmission. PARITY ERROR LED gives the indication.

Even Parity Mode

EVEN Parity Generator generates the coded sequence, which consists of 6 bits of data plus one EVEN Parity Bit for transmission. The least significant bit (B1) of the data word from the ADC is replaced with the parity bit. The coded sequence will be :Even Parity bit, B2, B3, B4, B5, B6, B7 (MSB)

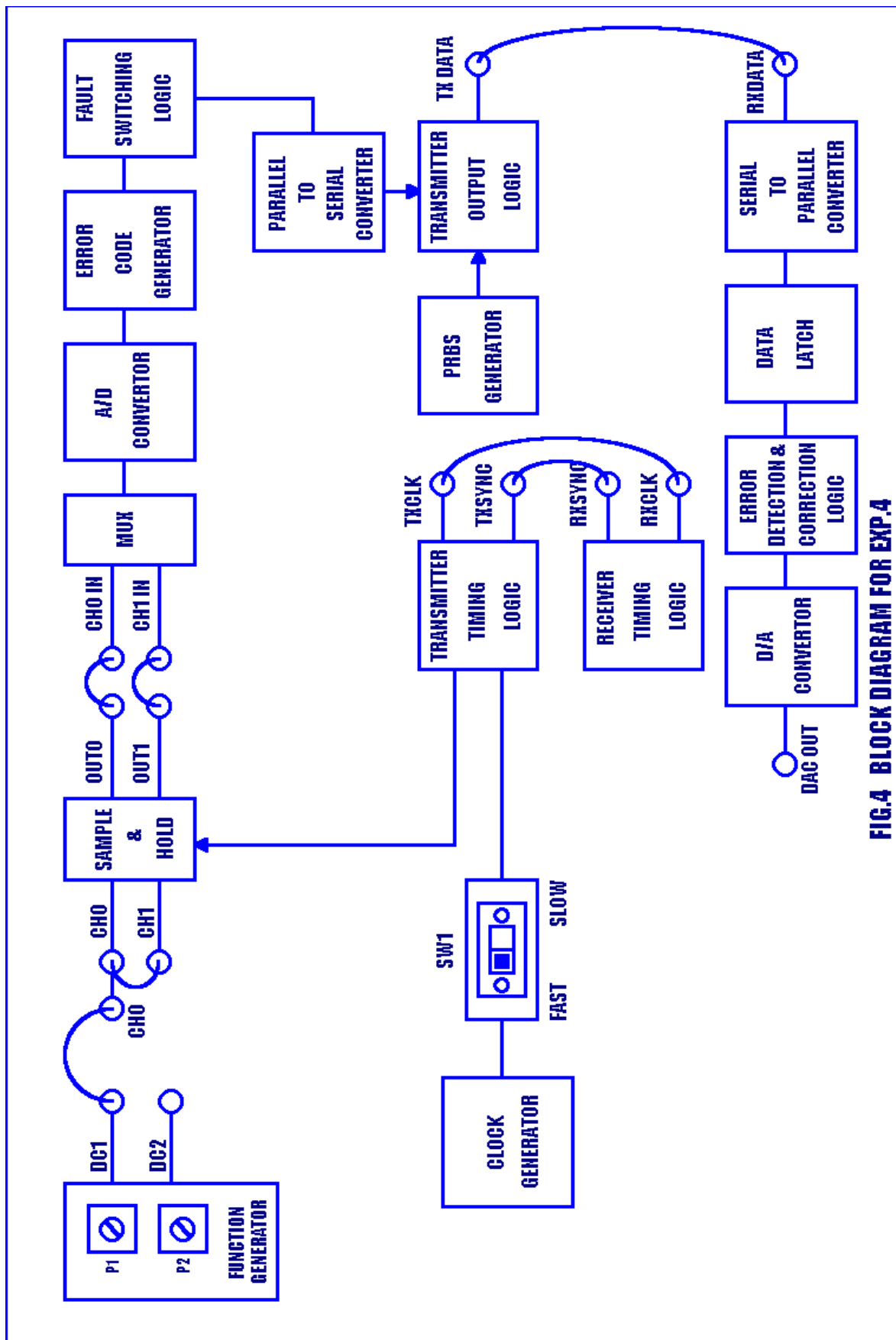
Even Parity Checker checks the number of Ones in the data stream and determines if a single bit error had occurred during transmission. PARITY ERROR LED gives the indication.

Hamming Parity Mode

Hamming Code Generator generates the coded sequence, which consists of 4 bits of data plus 3 bits of error Check Bits (K1, K2, K3) for the transmission. The 3 least significant bits (B1, B2, B3) of the data word from the ADC are neglected. The coded sequence will be: K1, K2, K3, B4, B5, B6, B7 (MSB) Hamming Error Detection/Correction Logic computes $K1^*$, $K2^*$, $K3^*$ at the receiver and locates the error bit position. Once the error bit position is located, the error is corrected by bit reversal

None Parity Mode

Operations without any error check is also available, where the transmitted data consists of 7 bits of data only. The coded sequence will be: B1, B2, B3, B4, B5, B6, B7 (MSB) Anyone of the above four modes of operations can be selected, by using the switches provided onboard.

**FIG.4 BLOCK DIAGRAM FOR EXP.4**

EQUIPMENTS:

- DCL-03 & DCL-04 Kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

PART A: None Parity Mode

1. Refer to the Block Diagram (Fig. 4) & Carry out the following connections.
2. Connect power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
3. Connect DC input signal **DC 1** to the input **CH 0** and **CH 1** of the Sample and Hold logic.
4. Set the speed selection switch **SW1** to **FAST** mode.
5. Select parity selection switch to **NONE** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
6. Connect **TXDATA**, **TXCLK** and **TXSYNC** of the transmitter section **DCL-03** to the corresponding **RXDATA**, **RXCLK**, and **RXSYNC** of the receiver section **DCL-04**.
7. Vary the amplitude of input DC signal from 0V to 4.96V and observe the variation on LED on the transmitter and receiver as mention below.
8. Create a single bit fault in any one of the 4 – MSB data bit by putting switch in below position of **SF 1** and observe the status of **PARITY ERROR**.
9. Repeat the experiment in **SLOW** mode.

OBSERVATIONS:

Observe the sequence of data bit on the LED for each setting and note down on the paper.

ON KIT **DCL-03**

- A/D Converter
- Parity Coded Data
- Error Code Generator

ON KIT **DCL-04**

- Shift Register
- Data Latch
- D/A Converter
- Parity Error

OBSERVATION TABLE:**None Parity Mode**

A/D Convertor	Error Code Generator	Shift Register (Fault)	Data Latch	D/A Convertor	Parity Error
0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	OFF
0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 1 0 0 0	0 0 0 1 0 0 0	OFF
1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	OFF

It has been observed that on creating a fault in the data stream there is no change in the status of Parity Check LED.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove connection between EX-OR gate output & multiplexer output in Parity generator logic. Due to this there is no Parity indication in B1 for PARITY CODED DATA.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **3** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will open the connection of seventh bit of Error Detection and correction logic. Due to this, fault inserted in B7 position is not recognized & not corrected in case of Hamming Parity mode.

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

PART B: Odd Parity Mode.

1. Refer to the Block Diagram (Fig. 4) & Carry out the following connections.
2. Connect power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
3. Connect DC input signal **DC 1** to the input **CH 0** and **CH 1** of the Sample and Hold logic.
4. Set the speed selection switch **SW1** to **FAST** mode.
5. Select parity selection switch to **ODD** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
6. Connect **TXDATA**, **TXCLK** and **TXSYNC** of the transmitter section **DCL-03** to the corresponding **RXDATA**, **RXCLK**, and **RXSYNC** of the receiver section **DCL-04**.
7. Vary the amplitude of input DC signal from 0V to 4.96V and observe the variation on LED on the transmitter and receiver as mention below.
8. Create a single bit fault in any one of the 4 – MSB data bit by putting switch in below position of **SF 1** and observe the status of **PARITY ERROR**.
9. Repeat the experiment in **SLOW** mode.

OBSERVATION:

Observe the sequence of data bit on LED for each setting and note down on the paper.

ON KIT **DCL-03**

- A/D Converter
- Parity Coded Data
- Error Code Generator

ON KIT **DCL-04**

- Shift Register
- Data Latch
- D/A Converter
- Parity Error Bit

OBSERVATION TABLE:

Odd Parity Mode

A/D CONVERTO R	ERROR CODE GENERATOR	SHIFT REGISTER (FAULT)	DATA LATCH	D/A CONVERTO R	PARIT Y ERRO R
0 0 0 0 0 0 0	1 0 0 0 0 0 0	1 0 0 0 0 0 0	1 0 0 0 0 0 0	0 0 0 0 0 0 0	OFF
0 0 0 0 0 0 0	1 0 0 0 0 0 0	1 0 0 1 0 0 0	1 0 0 1 0 0 0	0 0 0 1 0 0 0	ON
1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	0 1 1 1 1 1 1	OFF

We observe that the LSB of the A/D converter output is neglected in this mode of operations and the odd parity occupies the positions of LSB in transmission.

Also the LSB of the D/A Converter is always zeroing as in that position odd parity bit was transmitted.

Whenever the transmission of data is error free, the error checks LED remains OFF.

Whenever a single bit error occurs the parity check LED remains ON, indicating that a single bit error had occurred in transmission. Thus the Odd Parity check is able to detect the errors but unable to locate or correct the errors.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove connection between EX-OR gate output & multiplexer output in Parity generator logic. Due to this there is no Parity indication in B1 for PARITY CODED DATA.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **3** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will open the connection of seventh bit of Error Detection and correction logic. Due to this, fault inserted in B7 position is not recognized & not corrected in case of Hamming Parity mode.

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

PART C: Even Parity Mode

1. Refer to the Block Diagram (Fig. 4) & Carry out the following connections.
2. Connect power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
3. Connect DC input signal **DC 1** to the input **CH 0** and **CH 1** of the Sample and Hold logic.
4. Set the speed selection switch **SW1** to **FAST** mode.
5. Select parity selection switch to **EVEN** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).

6. Connect **TXDATA**, **TXCLK** and **TXSYNC** of the transmitter section **DCL-03** to the corresponding **RXDATA**, **RXCLK**, and **RXSYNC** of the receiver section **DCL-04**.
7. Vary the amplitude of input DC signal from 0V to 4.96V and observe the variation on LED on the transmitter and receiver as mention below.
8. Create a single bit fault in any one of the 4 – MSB data bit by putting switch in below position of **SF 1** and observe the status of **PARITY ERROR**.
9. Repeat the experiment in **SLOW** mode.

OBSERVATION:

Observe the sequence of data bit on LED for each setting and note down on the paper.

ON KIT DCL-03

- A/D Converter
- Parity Coded Data
- Error Code Generator

ON KIT DCL-04

- Shift Register
- Data Latch
- D/A Converter
- Parity Error Bit

OBSERVATION TABLE

Even Parity Mode

A/D CONVERTOR	ERROR CODE GENERATOR	SHIFT REGISTER (FAULT)	DATA LATCH	D/A CONVERTOR	PARITY ERROR
0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	OFF
0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 1 0 0 0	0 0 0 1 0 0 0	ON
1 1 1 1 1 1 1	0 1 1 1 1 1 1	0 1 1 1 1 1 1 1	0 1 1 1 1 1 1	0 1 1 1 1 1 1	OFF

We observe that the LSB of the A/D converter output is neglected in this mode of operations and the odd parity occupies the positions of LSB in transmission.

Also the LSB of the D/A Converter is always zeroing as in that position odd parity bit was transmitted.

Whenever the transmission of data is error free, the error check LED remains OFF.

Whenever a single bit error occurs the parity check LED remains ON indicating that a single bit error had occurred in transmission. Thus the Even Parity check is able to detect the errors but unable to locate or correct the errors.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove connection between EX-OR gate output & multiplexer output in Parity generator logic. Due to this there is no Parity indication in B1 for PARITY CODED DATA.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **3** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will open the connection of seventh bit of Error Detection and correction logic. Due to this, fault inserted in B7 position is not recognized & not corrected in case of Hamming Parity mode.

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

PART D: Hamming Parity Mode

1. Refer to the Block Diagram (Fig. 4) & Carry out the following connections.
2. Connect power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
3. Connect DC input signal **DC 1** to the input **CH 0** and **CH 1** of the Sample and Hold logic.
4. Set the speed selection switch **SW1** to **FAST** mode.
5. Select parity selection switch to **HAMMING** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
6. Connect **TXDATA**, **TXCLK** and **TXSYNC** of the transmitter section **DCL-03** to the corresponding **RXDATA**, **RXCLK**, and **RXSYNC** of the receiver section **DCL-04**.
7. Vary the amplitude of input DC signal from 0V to 4.96V and observe the variation on LED on the transmitter and receiver as mention below.
8. Create a single bit fault in any one of the 4 – MSB data bit by putting switch in below position of **SF 1** and observe the status of **PARITY ERROR**.
9. Repeat the experiment in **SLOW** mode.

OBSERVATION:

Observe the sequence of data bit on LED for each setting and note down on the paper.

ON KIT DCL-03

- A/D Converter
- Parity Coded Data
- Error Code Generator

ON KIT DCL-04

- Shift Register
- Data Latch
- D/A Converter
- Parity Error Bit

OBSERVATION TABLE:**Hamming Parity Mode**

A/D CONVERTOR	PARITY CODED DATA	ERROR CODE GENERATOR	DATA LATCH	D/A CONVERTOR	ERROR DETECTION/ CORRECTION
* * * 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0
* * * 0 0 0 0	0 0 0 0 0 0 0	0 0 0 1 0 0 0	0 0 0 1 0 0 0	0 0 0 0 0 0 0	0 0 0 1 0 0 0
* * * 1 0 0 0	1 1 0 1 0 0 0	1 1 0 1 0 0 0	1 1 0 1 0 0 0	0 0 0 1 0 0 0	0 0 0 0 0 0 0
* * * 0 1 0 0	1 0 1 0 1 0 0	1 0 1 0 1 0 0	1 0 1 0 1 0 0	0 0 0 0 1 0 0	0 0 0 0 0 0 0
* * * 1 1 0 0	0 1 1 1 1 0 0	0 1 1 1 1 0 0	0 1 1 1 1 0 0	0 0 0 1 1 0 0	0 0 0 0 0 0 0
* * * 0 0 1 0	0 1 1 0 0 1 0	0 1 1 0 0 1 0	0 1 1 0 0 1 0	0 0 0 0 0 1 0	0 0 0 0 0 0 0
* * * 1 0 1 0	1 0 1 1 0 1 0	1 0 1 1 0 1 0	1 0 1 1 0 1 0	0 0 0 1 0 1 0	0 0 0 0 0 0 0
* * * 0 1 1 0	1 1 0 0 1 1 0	1 1 0 0 1 1 0	1 1 0 0 1 1 0	0 0 0 0 1 1 0	0 0 0 0 0 0 0
* * * 1 1 1 0	0 0 0 1 1 1 0	0 0 0 1 1 1 0	0 0 0 1 1 1 0	0 0 0 1 1 1 0	0 0 0 0 0 0 0
* * * 0 0 0 1	1 1 1 0 0 0 1	1 1 1 0 0 0 1	1 1 1 0 0 0 1	0 0 0 0 0 0 1	0 0 0 0 0 0 0
* * * 1 0 0 1	0 0 1 1 0 0 1	0 0 1 1 0 0 1	0 0 1 1 0 0 1	0 0 0 1 0 0 1	0 0 0 0 0 0 0
* * * 0 1 0 1	0 1 0 0 1 0 1	0 1 0 0 1 0 1	0 1 0 0 1 0 1	0 0 0 0 1 0 1	0 0 0 0 0 0 0
* * * 1 1 0 1	1 0 0 1 1 0 1	1 0 0 1 1 0 1	1 0 0 1 1 0 1	0 0 0 1 1 0 1	0 0 0 0 0 0 0
* * * 0 0 1 1	1 0 0 0 0 1 1	1 0 0 0 0 1 1	1 0 0 0 0 1 1	0 0 0 0 0 1 1	0 0 0 0 0 0 0
* * * 1 0 1 1	0 1 0 1 0 1 1	0 1 0 1 0 1 1	0 1 0 1 0 1 1	0 0 0 1 0 1 1	0 0 0 0 0 0 0
* * * 0 1 1 1	0 0 1 0 1 1 1	0 0 1 0 1 1 1	0 0 1 0 1 1 1	0 0 0 0 1 1 1	0 0 0 0 0 0 0
* * * 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	0 0 0 1 1 1 1	0 0 0 0 0 0 0

We observe that the three LSB data bit of the A/D converter output is neglected in this mode of operations and three Hamming parity bit occupies the positions in transmission.

Also four MSB bit of the D/A converter forms the data and three LSB bit are always zero as in that position Hamming parity bit were transmitted.

Whenever the transmission of data is error free, all the LED of Error Detection/ Correction logic remains OFF.

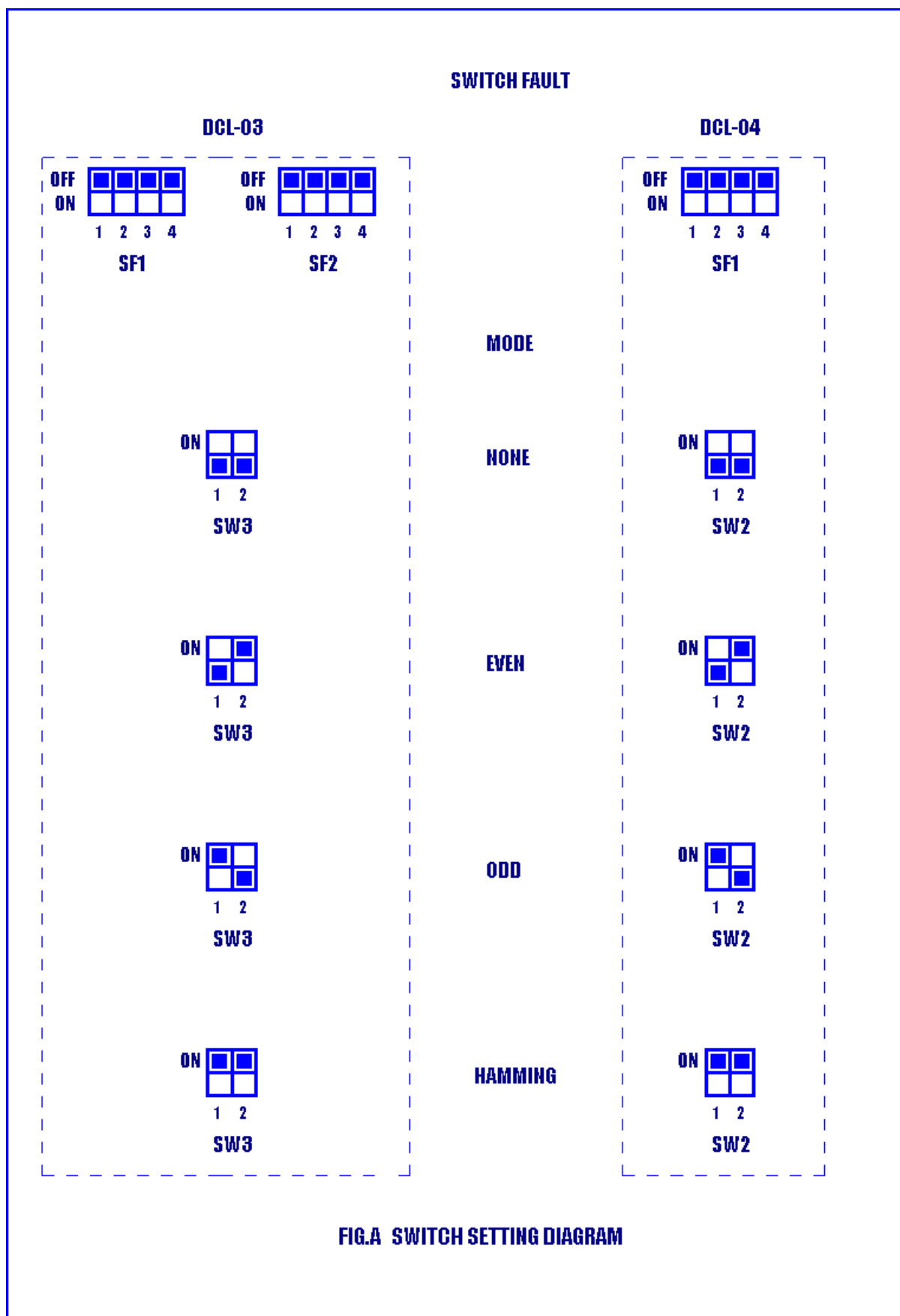
Whenever a single bit error occurs the corresponding bit position is indicated by LED of Error Detection/ Correction Logic and corrected data bit are observed at the input of D/A Converter which are same as the A/D Converter output (4 MSB) Thus in Hamming Parity mode, single bit error is detected as well as corrected.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove connection between EX-OR gate output & multiplexer output in Parity generator logic. Due to this there is no Parity indication in B1 for PARITY CODED DATA.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **3** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will open the connection of seventh bit of Error Detection and correction logic. Due to this, fault inserted in B7 position is not recognized & not corrected in case of Hamming Parity mode.

EXPERIMENT NO.5



EXPERIMENT NO: 5

NAME:

Synchronization Techniques

OBJECTIVE:

Study of Synchronization techniques

THEORY:

Synchronization Techniques

For the recovery of information from the transmitted data stream, the transmitter and receiver should be perfectly synchronized. Synchronization between transmitter and receiver is achieved in two stages:

- 01) Bit Synchronization; and
- 02) Frame Synchronization.

Bit Synchronization

To recover the information from the serial bit stream, the receiver has to know, if the received data bit at a given instant of time, is a ONE or a ZERO. For determining the data bit, the clocks of transmitter and receiver are to be perfectly synchronized. Once this Synchronization is achieved, the receiver recovers the data, with reference to the clock.

For achieving this Synchronization, normally known as bit Synchronization. the clock of the transmitter (TXCLK) can be sent along with the data. This calls for an additional channel for clock transmission. So normally a phase locked loop (PLL) is used at the Receiver for recovering the clock. The phase locked loop is made to lock with the incoming data whose VCO reproduces the clock at the Receiver.

Frame Synchronization

To recover the information transmitted from the data stream, the data is first converted to parallel code words. Then code words are decoded using the D/A converter, demultiplexing logic assigns the samples to the corresponding channel reconstruction units. For ensuring proper recovery of code words, and proper assignment of channel information of the demultiplexing unit, the receiver has to know where the data begins exactly. For this, at the start of every frame, a 14-bit pseudo random sequence (PRBS) is sent, which forms the header of the data pattern. Subsequently what follows is data corresponding to the number of channels multiplexed.

The frame sync pulse can be sent by the Transmitter along with the data and clock to achieve frame Synchronization. But normally it is not sent, since it calls for an additional channel.

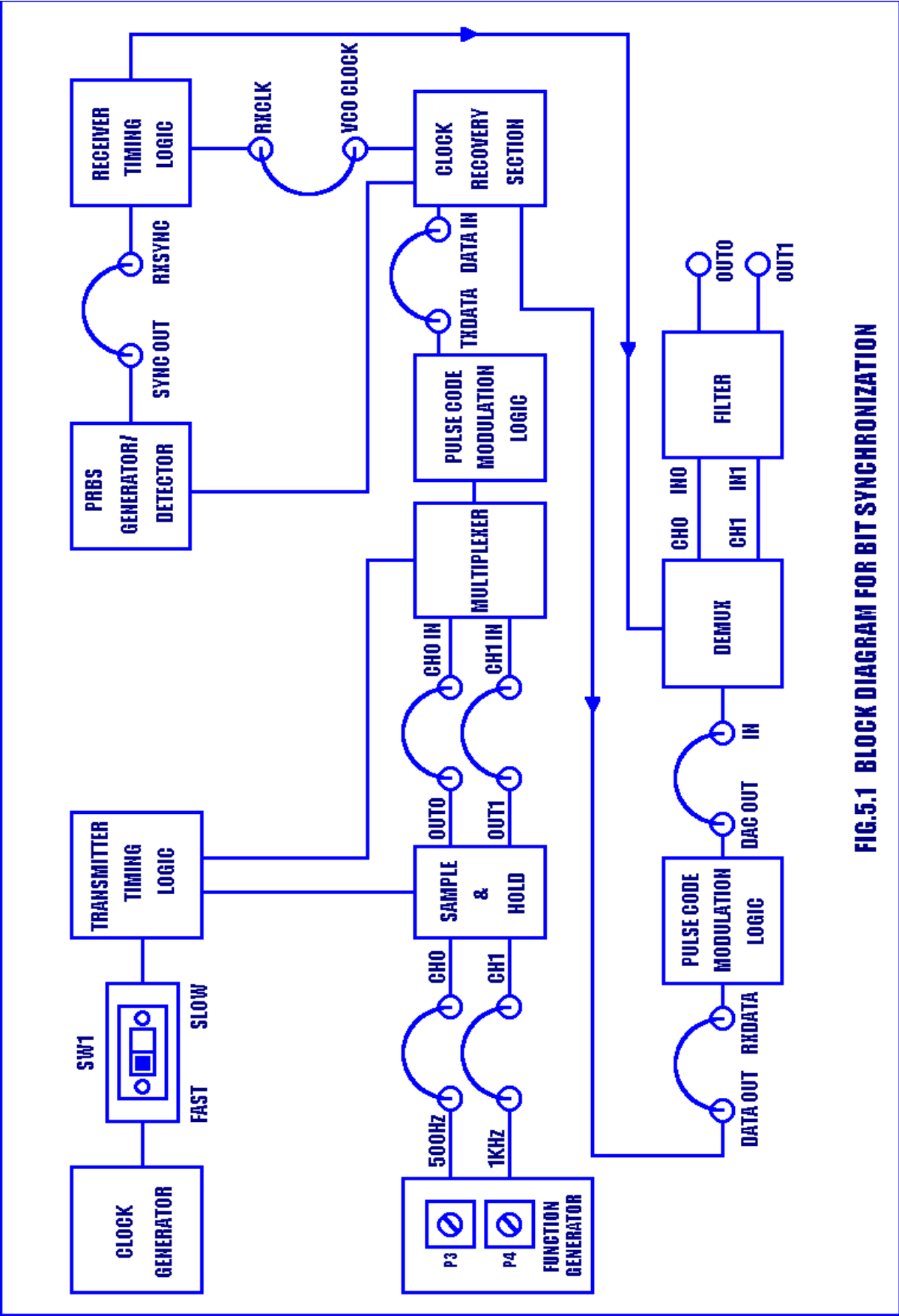


FIG.5.1 BLOCK DIAGRAM FOR BIT SYNCHRONIZATION

To recover the frame sync at the receiver from the data stream, a sequence. To Recover the frame sync at the receiver from the data stream, a sequence similar to the one transmitted, is generated at the Receiver. The PRBS detector compares the generated sequence with the incoming data and locks whenever the incoming data matches with the generated sequence. The PRBS sequence is chosen in such a way that it have minimum correlation with the data to avoid false locks of the detector with the data bits. Once the PRBS detector is locked, it generates a frame sync pulse, whose rising edge corresponds to the start of the frame and the time to the slot when the pseudo random sequence occurs in the data stream. This frame sync pulse is used to code word and for demultiplexing purposes.

EQUIPMENT:

- DCL-03 & DCL-04 Kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

I. Bit Synchronization

1. Refer to the Block Diagram (Fig. 5.1) & Carry out the following connections.
2. Connect the power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
3. Connect sine wave of frequency **500Hz** and **1 KHz** to the input **CH0** and **CH1** of the sample and hold logic.
4. Connect **OUT 0** to **CH0 IN** & **OUT 1** to **CH1 IN**.
5. Select the speed selection switch **SW1** to **FAST** mode.
6. Select parity selection switch to **NONE** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
7. Connect **TXDATA** of **DCL-03** to the **DATA IN** of the receiver section **DCL-04**.
8. Ensure that **FAULT SWITCH SF 1** introduces no fault as shown in switch setting diagram (Fig. A).
9. Connect **DATAOUT** to **RXDATA**, **VCO CLOCK** post to **RXCLK** and **SYNCOUT** to the **RXSYNC** with the help of connecting chords in **DCL-04** Kit.
10. Connect **DAC OUT** to **IN** posts on **DCL-04**
11. Take the observations as mentioned below.
12. Connect ground points of both the kits with the help of connecting chord provided during the all experiments.

II. Frame Synchronization

1. Refer to the Block Diagram (Fig. 5.2) & Carry out the following connections.
2. Connect the power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
3. Connect sine wave of frequency **500Hz** and **1 KHz** to the input **CH0** and **CH1** of the sample and hold logic.
4. Connect **OUT 0** to **CH0 IN** & **OUT 1** to **CH1 IN**.
5. Select the speed selection switch **SW1** to **FAST** mode.

6. Select parity selection switch to **NONE** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
7. Connect **TXDATA** of **DCL-03** to **RXDATA** of the receiver section of **DCL-04**.
8. Ensure that **FAULT SWICH SF1** as shown in switch setting diagram (Fig. A) introduces no fault.
9. Connect the **TXCLK** of **DCL-03** to the **RXCLK** of **DCL-04**.
10. Connect the **SYNCOUT** of **DCL-04** to **RXSYNC** of **DCL-04**.
11. Connect **DAC OUT** to **IN** posts on **DCL-04**
12. Take the observation as mentioned below.
13. Connect ground points of both the kits with the help of connecting chord provided during all the experiments.

OBSERVATION:

Observe the following signal on CRO and plot it on the paper.

ON KIT DCL-03

- Input signal **CH 0** and **CH 1**.
- Sample and Hold output **OUT 0** and **OUT1**
- Multiplexer clock **CLK 1** and **CLK 2**
- Multiplexed data **MUX OUT**.
- PCM Data **TX DATA**, **TXCLK**, **TXSYNC**

ON KIT DCL-04

- **RXCLK**, **RXSYNC**, **RXDATA**
- **VCO CLOCK**, **SYNCOUT**
- Demultiplexer clock **CLK 1** and **CLK 2**
- Demultiplexed Data **CH 0** and **CH 1**
- Received signal **OUT 0** and **OUT 1**

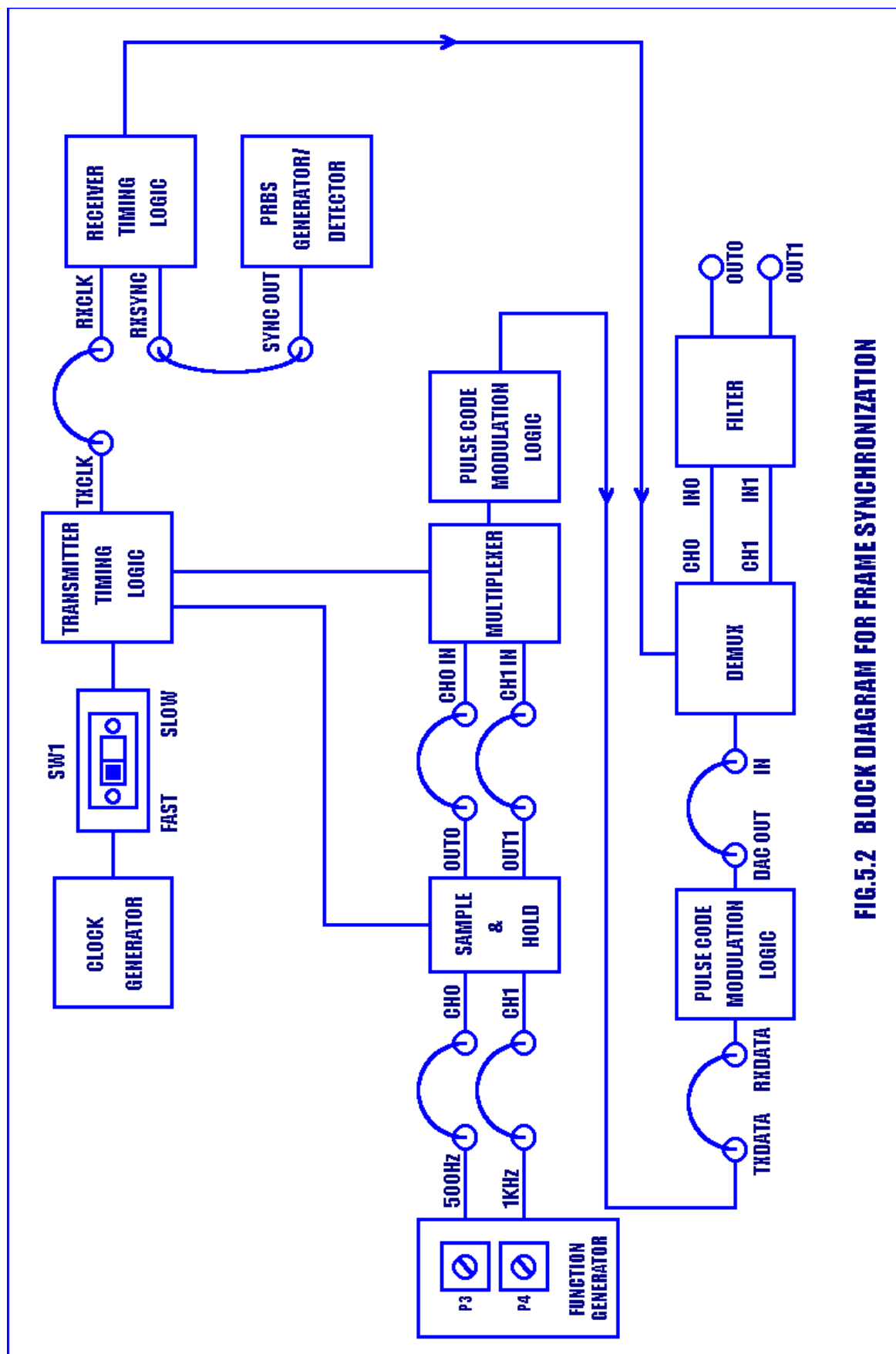


FIG.5.2 BLOCK DIAGRAM FOR FRAME SYNCHRONIZATION

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **2** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove PRBS sequence from Transmitted Data. Synchronization will be only possible in case of direct connection of TXSYNC to RXSYNC. In any other case no synchronization is possible.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **4** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will remove CH0 filter capacitor connection from ground. Filter output will be distorted.

CONCLUSION:

The PLL will lock onto the frequency of the generated sync pulse and the PLL along with the associated circuitry will generate the sync and clock signals, which are same in phase and frequency as that of the TXCLK and TXSYNC. This enables the receiver to get the output.

EXPERIMENT NO.6

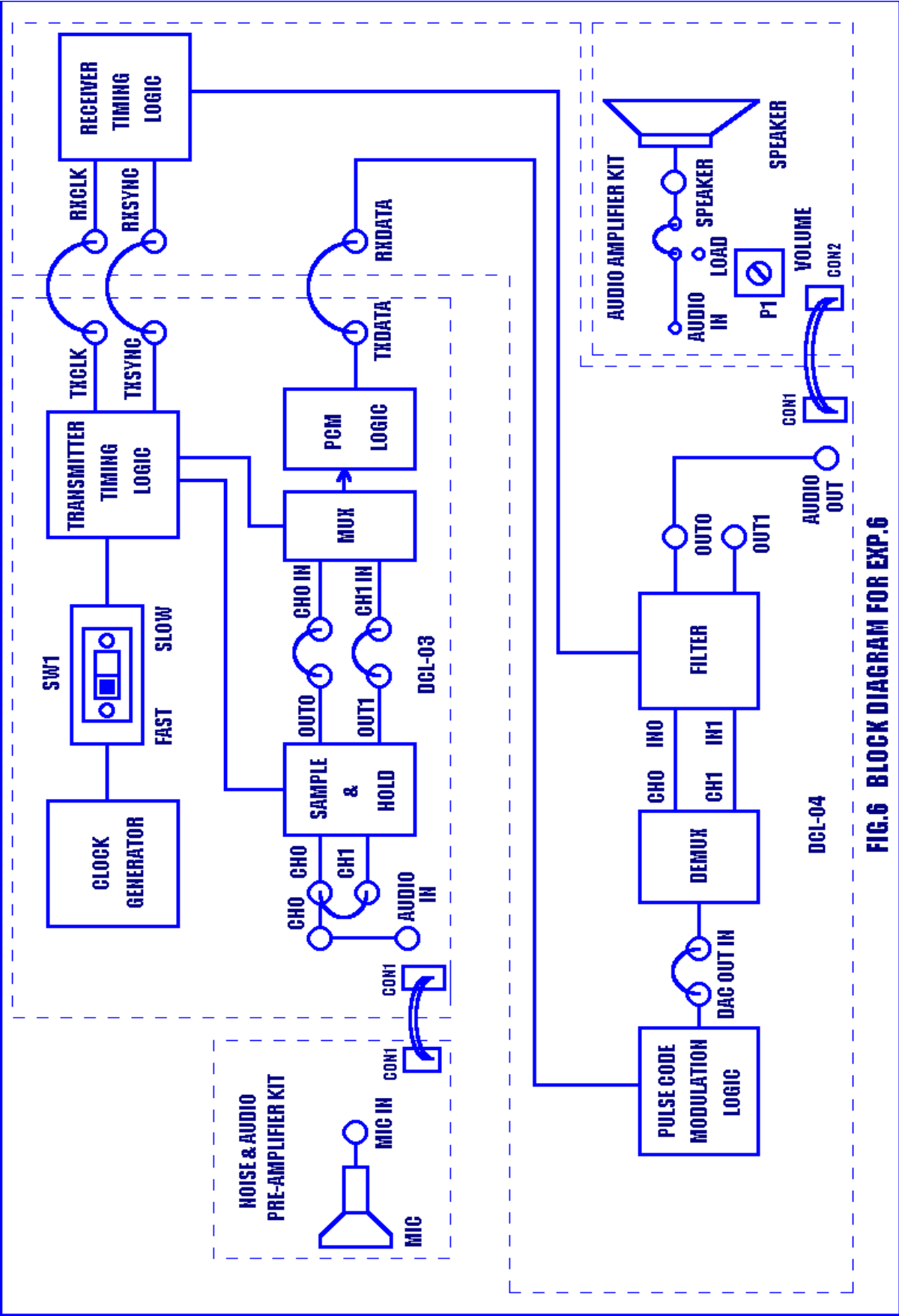


FIG. 6 BLOCK DIAGRAM FOR EXP. 6

EXPERIMENT NO: 6

NAME:

Voice Communication

OBJECTIVE:

The objective of this experiment is to study the voice communication using DCL-03 & DCL-04 kit.

THEORY:

Voice can be modulated, demodulated and reconstructed using any of the modulation techniques such as PAM, PWM, PPM & PCM. In DCL-03 kit Voice is converted to seven bit digital signal. It is then multiplexed with other data and sent serially to DCL-04 kit. This signal is then demodulated to recover the original voice signal at the Speaker.

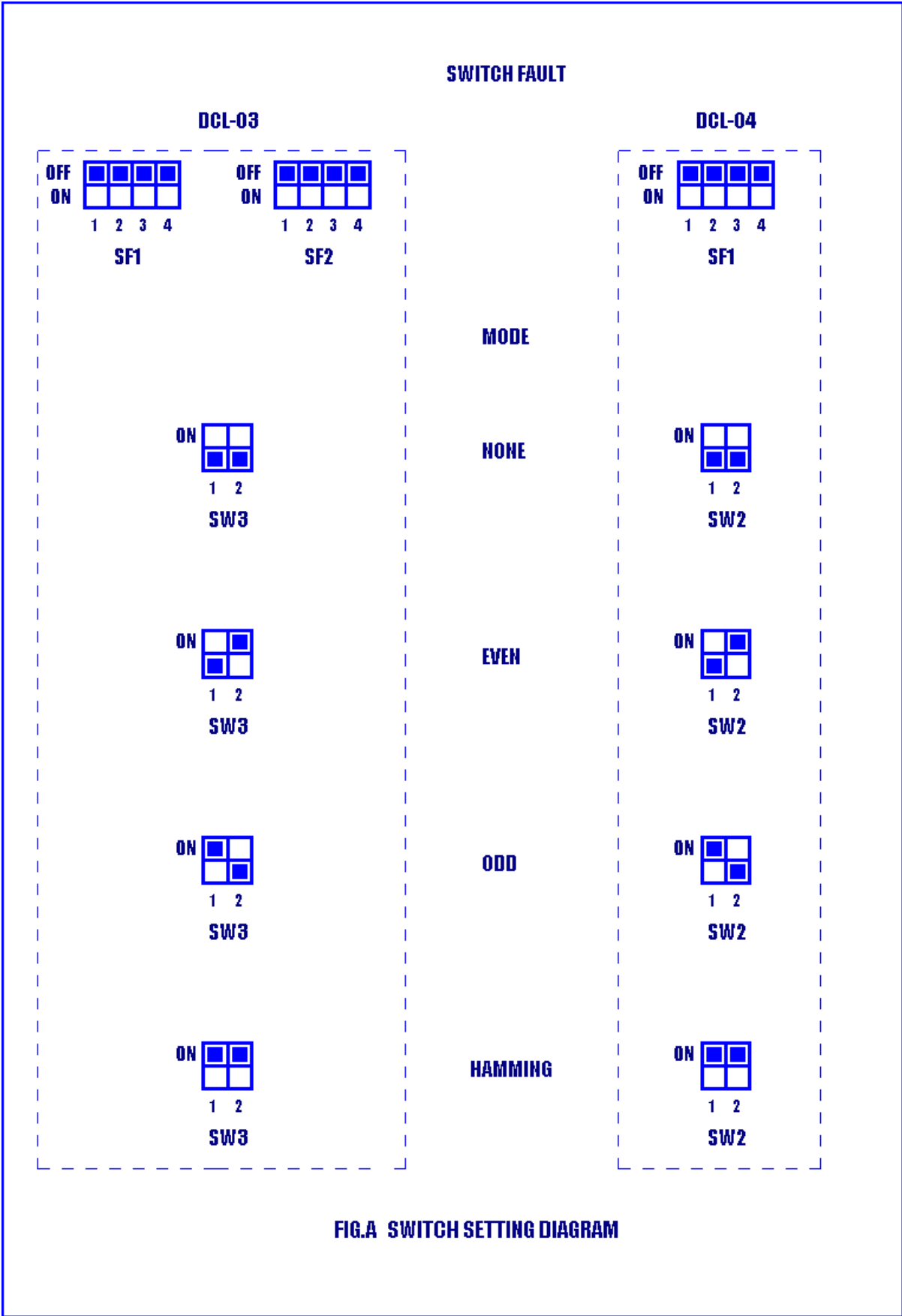
EQUIPMENTS:

- DCL-03 & DCL-04 Kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope
- Noise and Audio Pre-Amplifier kit
- Audio Amplifier kit
- Power connection cables
- Microphone

NOTE: Keep The Switch Faults In Off Position.

PROCEDURE:

1. Refer to the Block Diagram (Fig. 6) & Carry out the following connections.
2. Connect the Noise and Audio Pre-Amplifier kit to the **CON1** Post on the **DCL-03** kit using power connection cable.
3. Connect the Audio Amplifier kit to the **CON1** Post on the **DCL-04** kit using power connection cable.
4. Connect the power supply in proper polarity to the kits **DCL-03** and **DCL-04** and switch it on.
5. Connect Microphone to the post **MIC IN** on Noise and Audio Pre-Amplifier kit.



6. Connect the following posts on **DCL-03**
AUDIO IN to **CH 0**
CH 0 to **CH 1**
OUT 0 to **CH0 IN**
OUT 1 to **CH1 IN**
Connect the following posts on **DCL-04**
DAC OUT to **IN**
OUT 1 / OUT 2 to **AUDIO OUT**
7. Select the speed selection switch **SW1** to **FAST** mode.
8. Select parity selection switch to **NONE** mode on both the kit **DCL-03** and **DCL-04** as shown in switch setting diagram (Fig. A).
9. Connect **TXDATA**, **TXCLK** and **TXSYNC** of the transmitter section **DCL-03** to the corresponding **RXDATA**, **RXCLK**, and **RXSYNC** of the receiver section **DCL-04**.
10. Observe the waveforms for voice signal, PCM of voice signal.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **2** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will remove PRBS sequence from Transmitted Data. Synchronization will be only possible in case of direct connection of TXSYNC to RXSYNC. In any other cases no synchronization is possible.
- Put switch **3** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will increase the duty cycle (on period) of CH0. Due to which, some portion of CH0 signal present in CH1 signals time slot. Hence demultiplexed output for CH1 is disturbed.
- Put switch **4** of **SF2 (DCL-03)** in Switch Fault section to ON position. This will disable Data from going to TX Data, only PRBS will be present.
- Put switch **1** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable Synch signal in Bit Synchronization. Filter output will be disturbed.
- Put switch **2** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will disable clock signal for Demultiplexer. One channel output is absent and other channel output is disturbed.
- Put switch **4** of **SF1 (DCL-04)** in Switch Fault section to ON position. This will remove CH0 filter capacitor connection from ground. Filter output will be distorted.

CONCLUSION:

We will get clear voice communication when faults are not introduced in the circuit. When we introduce faults in bit positions, detected signal is distorted and hence voice also gets disturbed.